JESS Thermodynamic Database v8.9

Boron

24-Jan-23

Reaction No. 2522							
Fe+3_B(OH)3 + B(OH)3 = Fe+3_B(OH)3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:7.10(3SF)	0	MSC	22560
2	23	0	Unknown	lgK:6.96(3SF)	0	MSC	22560
3	25	0.68	Unknown	lgK:1.0(2SF)	2	MSP	413
Note (3): Given in IUPAC database [05PeP_22560]							

Reaction No. 2569							
B(OH)4-1 + H+1 = B(OH)3 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0	P=100 Inf. Dilution	lgK:9.350(4SF)	5	MDG	6380
2	25	0	P=100 Inf. Dilution	lgK:9.145(4SF)	5	MDG	6380
3	50	0	P=100 Inf. Dilution	lgK:9.150(4SF)	5	MDG	6380
4	0	0	P=200 Inf. Dilution	lgK:9.277(4SF)	5	MDG	6380
5	25	0	P=200 Inf. Dilution	lgK:9.084(4SF)	5	MDG	6380
6	50	0	P=200 Inf. Dilution	lgK:9.093(4SF)	5	MDG	6380
7	0	0	P=300 Inf. Dilution	lgK:9.205(4SF)	5	MDG	6380
8	25	0	P=300 Inf. Dilution	lgK:9.025(4SF)	5	MDG	6380
9	50	0	P=300 Inf. Dilution	lgK:9.037(4SF)	5	MDG	6380
10	0	0	P=400 Inf. Dilution	lgK:9.135(4SF)	5	MDG	6380
11	25	0	P=400 Inf. Dilution	lgK:8.966(4SF)	5	MDG	6380
12	50	0	P=400 Inf. Dilution	lgK:8.982(4SF)	5	MDG	6380
13	0	0	P=500 Inf. Dilution	lgK:9.066(4SF)	5	MDG	6380

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14	22	0	P=500 Inf. Dilution	lgK:8.926(4SF)	5	MDG	24193
Note (14): Calc from lgK0=9.236 & KP/K0 = 2.04							
15	25	0	P=500 Inf. Dilution	lgK:8.909(4SF)	5	MDG	6380
16	50	0	P=500 Inf. Dilution	lgK:8.929(4SF)	5	MDG	6380
17	0	0	P=600 Inf. Dilution	lgK:8.999(4SF)	5	MDG	6380
18	25	0	P=600 Inf. Dilution	lgK:8.853(4SF)	5	MDG	6380
19	50	0	P=600 Inf. Dilution	lgK:8.876(4SF)	5	MDG	6380
20	0	0	P=700 Inf. Dilution	lgK:8.933(4SF)	5	MDG	6380
21	25	0	P=700 Inf. Dilution	lgK:8.798(4SF)	5	MDG	6380
22	50	0	P=700 Inf. Dilution	lgK:8.825(4SF)	5	MDG	6380
23	0	0	P=800 Inf. Dilution	lgK:8.868(4SF)	5	MDG	6380
24	25	0	P=800 Inf. Dilution	lgK:8.745(4SF)	5	MDG	6380
25	50	0	P=800 Inf. Dilution	lgK:8.775(4SF)	5	MDG	6380
26	0	0	P=900 Inf. Dilution	lgK:8.805(4SF)	5	MDG	6380
27	25	0	P=900 Inf. Dilution	lgK:8.693(4SF)	5	MDG	6380
28	50	0	P=900 Inf. Dilution	lgK:8.726(4SF)	5	MDG	6380
29	25	0	P=971 Inf. Dilution	lgK:8.680(4SF)	5	MDG	24193
Note (29): Calc from lgK0=9.236 & KP/K0 = 3.60							
30	0	0	P=1000 Inf. Dilution	lgK:8.743(4SF)	5	MDG	6380
31	22	0	P=1000 Inf. Dilution	lgK:8.656(4SF)	5	MDG	24193
Note (31): Calc from lgK0=9.236 & KP/K0 = 3.80							
32	25	0	P=1000 Inf. Dilution	lgK:8.642(4SF)	5	MDG	6380
33	50	0	P=1000 Inf. Dilution	lgK:8.678(4SF)	5	MDG	6380
34	25	0	P=1942 Inf. Dilution	lgK:8.252(4SF)	5	MDG	24193

Note (34): Calc from lgK0=9.236 & KP/K0 = 9.63							
35	5	0	Inf. Dilution/m	lgK:9.440(4SF)	5	MEF	422
36	10	0	Inf. Dilution	lgK:9.4(0.05SD)	5	MIS	420
37	10	0	Inf. Dilution	lgK:9.380(4SF)	4	CRV	552
38	10	0	Inf. Dilution/m	lgK:9.380(4SF)	5	MEF	422
39	15	0	Inf. Dilution/m	lgK:9.326(4SF)	5	MEF	422
40	18	0	Inf. Dilution	lgK:8.77(3SF)	0	ENS	7694
41	20	0	Inf. Dilution/m	lgK:9.279(4SF)	5	MEF	422
42	25	0	Inf. Dilution	lgK:9.23(3SF)	4	EOD	288
43	25	0	Inf. Dilution	lgK:9.24(3SF)	8	CRV	406
44	25	0	Inf. Dilution	lgK:9.24(3SF)	6	MHY	409
45	25	0	Inf. Dilution	lgK:9.236(0.05SD)	5	CMN	420
46	25	0	Inf. Dilution	lgK:9.11(3SF)	0	MHY	423
47	25	0	Inf. Dilution	lgK:9.236(4SF)	4	CRV	552
48	25	0	Inf. Dilution	lgK:9.24(3SF)	4	ENS	6405
49	25	0	Inf. Dilution	lgK:9.24(3SF)	5	ENS	6405
Note (49): p. 161							
50	25	0	Inf. Dilution	lgK:9.236(0.003SD)	4	CRV	6478
51	25	0	Inf. Dilution	lgK:8.64(3SF)	0	ENS	7694
52	25	0	Inf. Dilution	lgK:9.23(3SF)	4	CRV	8749
53	25	0	Inf. Dilution	lgK:9.0(2SF)	0	CRV	22551
54	25	0	CHEMVAL4	lgK:9.24(0.01SD)	6	ENS	5156
55	25	0	Inf. Dilution/m	lgK:9.237(4SF)	5	MEF	422
56	25	0	UCT_Sea	lgK:9.24(0.01SD)	4	EDH	5390
57	25	0.1	NaClO4	lgK:9.0(0.05SD)	5	MDM	423
58	25	0.1	NaClO4	lgK:8.98(3SF)	4	MHY	427
59	25	0.1	UCT_Sea	lgK:8.97(0.01SD)	4	EDH	5390
60	25	0.1	Unknown	lgK:9.22(3SF)	0	MHY	409
61	25	0.1	Unknown	lgK:9.0(0.03SD)	6	CRV	552
62	25	0.2	UCT_Sea	lgK:8.93(0.01SD)	4	EDH	5390
63	25	0.3	UCT_Sea	lgK:8.90(0.01SD)	4	EDH	5390
64	25	0.4	UCT_Sea	lgK:8.87(0.01SD)	4	EDH	5390
65	25	0.5	Na+1 salt	lgK:8.87(0.01SD)	6	CRV	552
66	25	0.5	UCT_Sea	lgK:8.86(0.01SD)	4	EDH	5390
67	25	0.5	Unknown	lgK:8.97(3SF)	5	ENS	6405
Note (67): p. 161							

68	25	0.6	UCT_Sea	lgK:8.85(0.01SD)	4	EDH	5390
69	25	0.7	K+1 salt	lgK:9.0(0.04SD)	0	CRV	552
70	25	0.7	KCl	lgK:9.0(0.03SD)	0	MPS	769
71	25	0.7	Li+1 salt	lgK:8.75(3SF)	6	CRV	552
72	25	0.7	Na+1 salt	lgK:8.85(0.02SD)	4	CRV	552
73	25	0.7	NaCl	lgK:8.9(0.05SD)	3	MGL	405
74	25	0.7	UCT_Sea	lgK:8.85(0.01SD)	4	EDH	5390
75	25	1	Na+1 salt	lgK:8.83(0.02SD)	3	CRV	552
76	25	1	Unknown	lgK:8.89(3SF)	6	CRV	552
77	25	2	KCl	lgK:9.0(0.05SD)	0	MCL	5438
78	25	2	NaCl	lgK:8.79(3SF)	2	CRV	552
79	25	2	Unknown	lgK:9.00(3SF)	3	CRV	552
80	25	3	KBr	lgK:9.1(0.05SD)	0	CRV	552
81	25	3	NaCl	lgK:8.81(0.02SD)	2	CRV	552
82	25	3	NaClO4	lgK:8.90(3SF)	3	MHY	423
83	25	3	NaClO4	lgK:9.00(0.05SD)	0	CMN	426
84	25	3	NaClO4	lgK:9.00(0.018MD)	2	MHY	427
85	25	3	NaClO4	lgK:8.9(0.05SD)	3	CRV	552
86	25	4	NaCl	lgK:8.91(3SF)	4	CRV	552
87	25	5	NaCl	lgK:9.02(3SF)	6	CRV	552
88	30	0	Inf. Dilution/m	lgK:9.198(4SF)	5	MEF	422
89	35	0	Inf. Dilution	lgK:9.16(3SF)	4	CRV	8749
90	35	0	Inf. Dilution/m	lgK:9.164(4SF)	5	MEF	422
91	40	0	Inf. Dilution	lgK:9.125(4SF)	4	CRV	552
92	40	0	Inf. Dilution/m	lgK:9.132(4SF)	5	MEF	422
93	50	0	Inf. Dilution	lgK:9.1(0.05SD)	5	MIS	420
94	50	0	Inf. Dilution	lgK:9.08(3SF)	4	CRV	8749
95	10	0	Inf. Dilution	dH:-3.9(0.1SD)	2	MIS	420
96	10	0	Inf. Dilution	dH:-3.9(2SF)	2	CRV	552
97	25	0	Inf. Dilution	dH:-3.4(0.1SD)	5	MIS	420
98	25	0	Inf. Dilution	dH:-3.2(2SF)	4	CRV	552
99	25	0	Inf. Dilution	dH:-3.22(0.1SD)	4	CRV	6478
100	25	1	Na+1 salt	dH:-3.5(2SF)	4	CRV	552
101	25	2	KCl	dH:-3.9(0.1SD)	5	MCL	5438
102	25	2	NaCl	dH:-4.0(2SF)	6	CRV	552
103	40	0	Inf. Dilution	dH:-2.6(2SF)	0	CRV	552

104	50	0	Inf. Dilution	dH:-2.0(0.1SD)	0	MIS	420

Reaction No. 2570							
B(OH)4-1 + B(OH)3 + H2O = H+1_B(OH)4-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.07(1SF)	3	CRV	552
2	25	1	KCl	lgK:-0.17(2SF)	6	CRV	552
3	25	3	NaClO4	lgK:0.7(0.05SD)	4	MIS	409
4	25	3	NaClO4	lgK:0.67(2SF)	6	CRV	552
5	25	1	KCl	dH:-1.1(2SF)	6	CRV	552

Reaction No. 2571							
H+1_B(OH)4-1(2) + B(OH)3 + H2O = H+1(2)_B(OH)4-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.00(3SF)	4	CRV	552
2	25	1	KCl	lgK:2.10(3SF)	6	CRV	552
3	25	3	NaClO4	lgK:1.4(0.05SD)	5	MIS	409
4	25	3	NaClO4	lgK:1.37(3SF)	6	CRV	552
5	25	1	KCl	dH:5.2(2SF)	6	CRV	552

Reaction No. 9590							
B(OH)3 + H+1(3)_Pyrogallol-3 = H+1(2)_B(OH)3_Pyrogallol-3_H2O(-1) + H2O + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:-6.40(0.01SD)	6	MGL	1090

Reaction No. 9591							
B(OH)3 + 2<H+1(3)_Pyrogallol-3> = H+1(5)_B(OH)3_Pyrogallol-3(2)_H2O(-2) + 2<H2O > + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:-2.80(0.01SD)	6	MGL	1090

Reaction No. 10483							
B(OH)3 + H+1_Thioglycerol-1 = B(OH)3_Thioglycerol-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:-7.79(3SF)	6	MGL	999

Reaction No. 10484
B(OH)3 + 2<H+1_Thioglycerol-1> = H+1_B(OH)3_Thioglycerol-1(2) + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:-6.12(3SF)	6	MGL	999

Reaction No. 14135

B(OH)4-1 + 4<H2O> + 8<e-1> = BH4-1 + 8<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.241(4SF)	3	CRV	7761
2	25	0	Inf. Dilution	dH:695.6(4SF)kJ	3	CRV	7761

Reaction No. 23380

B(OH)3 + H+1_SulfSal-3 = B+3_OH-1(2)_SulfSal-3 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.3	KNO3	lgK:0.98(2SF)	6	MGL	5987
2	25	0	Inf. Dilution	lgK:1.0(2SF)	1	ESC	
3	25			lgK:0.85(2SF)	0	MGL	5987

Reaction No. 28567

B(OH)3 + 2<DRibose> = B+3_H+1(-4)_DRibose(2) + H+1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-1.90(3SF)	4	CRV	2556

Reaction No. 28571

B(OH)3 + 2<DLyxose> = B+3_H+1(-4)_DLyxose(2) + H+1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-4.28(3SF)	4	CRV	2556

Reaction No. 28574

B(OH)3 + 2<DXylose> = B+3_H+1(-4)_DXylose(2) + H+1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-5.09(3SF)	4	CRV	2556

Reaction No. 28578

B(OH)3 + 2<DArabinose> = B+3_H+1(-4)_DArabinose(2) + H+1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-5.82(3SF)	4	CRV	2556

Reaction No. 28582

B(OH)3 + 2<LArabinose> = B+3_H+1(-4)_LArabinose(2) + H+1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-5.55(3SF)	4	CRV	2556

Reaction No. 28586

B(OH)3 + DFructose = B+3_H+1(-2)_DFructose_OH-1(2) + H+1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-5.9(2SF)	4	CRV	2556

Reaction No. 28587

B(OH)3 + 2<DFructose> = B+3_H+1(-4)_DFructose(2) + H+1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-4.00(3SF)	4	CRV	2556

Reaction No. 28590

B(OH)3 + 2<LSorbose> = B+3_H+1(-4)_LSorbose(2) + H+1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-3.30(3SF)	4	CRV	2556

Reaction No. 28595

B(OH)3 + DGlucose = B+3_H+1(-2)_DGlucose_OH-1(2) + H+1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-6.95(3SF)	4	CRV	2556

Reaction No. 28596

B(OH)3 + 2<DGlucose> = B+3_H+1(-4)_DGlucose(2) + H+1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-6.34(3SF)	4	CRV	2556

Reaction No. 28600

B(OH)3 + 2<DMannose> = B+3_H+1(-4)_DMannose(2) + H+1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-4.58(3SF)	4	CRV	2556

Reaction No. 28605

B(OH)3 + 2<DGalactose> = B+3_H+1(-4)_DGalactose(2) + H+1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.9(2SF)	1	ELC	
2	25	0.1	Unknown	lgK:-6.79(3SF)	0	CRV	2556

Reaction No. 32626

B(s) + 2<F2(g)> + e-1 = BF4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-355.50(5SF)	3	CRV	406
2	25	0	Inf. Dilution	dG:-355.4(4SF)	4	CRV	6488
3	25	0	Inf. Dilution	dG:-1487.(4SF)kJ	5	CRV	7421
4	25	0	Inf. Dilution	dG:-355.40(5SF)	4	CRV	10174
5	50	0	Inf. Dilution	dG:-356.47(5SF)	0	CRV	10174
6	75	0	Inf. Dilution	dG:-357.54(5SF)	0	CRV	10174
7	100	0	Inf. Dilution	dG:-358.60(5SF)	0	CRV	10174
8	125	0	Inf. Dilution	dG:-359.65(5SF)	0	CRV	10174
9	150	0	Inf. Dilution	dG:-360.69(5SF)	0	CRV	10174
10	175	0	Inf. Dilution	dG:-361.71(5SF)	0	CRV	10174
11	200	0	Inf. Dilution	dG:-362.71(5SF)	0	CRV	10174
12	225	0	Inf. Dilution	dG:-363.68(5SF)	0	CRV	10174
13	250	0	Inf. Dilution	dG:-364.59(5SF)	0	CRV	10174
14	300	0	Inf. Dilution	dG:-366.14(5SF)	0	CRV	10174
15	25	0	Inf. Dilution	dH:-376.4(4SF)	4	CRV	6488

Reaction No. 33176

B(s) + 1.5<F2(g)> + 0.5<H2(g)> + 0.5<O2(g)> + e-1 = B(OH)F3-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-338.18(5SF)	5	CRV	406

Reaction No. 33580

B(OH)3 + H+1(3)_Gallic-4 = H+1 + B+3_H+1_OH-1(2)_Gallic-4 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-4.86(3SF)	5	CRV	2556

Reaction No. 33825

B(OH)3 + H+1(3)_Dopamine-2 = H+1 + B+3_H+1_OH-1(2)_Dopamine-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:-4.862(4SF)	5	MGL	3052

Reaction No. 33827

B(OH)3 + H+1(3)_Isoprenaline-2 = H+1 + H2O + B+3_H+1_OH-1(2)_Isoprenaline-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:-4.809(4SF)	5	MGL	3063

Reaction No. 34382

H+1 + 112COODodecBorane-2 = H+1_112COODodecBorane-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.23(4SF)	5	ENS	774
2	25	0	Inf. Dilution	dH:-2.3(2SF)	5	MCL	774

Reaction No. 34383

H+1_112COODodecBorane-2 + H+1 = H+1(2)_112COODodecBorane-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.07(3SF)	5	ENS	774
2	25	0	Inf. Dilution	dH:-2.15(3SF)	5	MCL	774

Reaction No. 34718

B(OH)4-1 + 2<DFructose> = B+3_H+1(-4)_DFructose(2) + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.1	KCl	lgK:5.109(4SF)	5	MGL	931
2	15	0.1	KCl	lgK:5.062(4SF)	5	MGL	931
3	25	0.1	KCl	lgK:5.04(3SF)	5	MGL	931
4	25	0.1	KCl	lgK:4.917(4SF)	5	MGL	931
5	35	0.1	KCl	lgK:4.772(4SF)	5	MGL	931
6	45	0.1	KCl	lgK:4.643(4SF)	5	MGL	931
7	25	0.1	KCl	dH:-5.9(2SF)	5	ENS	931

Reaction No. 34719

B(OH)4-1 + DFructose = B+3_H+1(-2)_DFructose_OH-1(2) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.1	KCl	lgK:4.142(4SF)	5	MGL	931
2	15	0.1	KCl	lgK:3.642(4SF)	5	MGL	931
3	25	0.1	KCl	lgK:3.416(4SF)	5	MGL	931
4	35	0.1	KCl	lgK:3.178(4SF)	5	MGL	931
5	45	0.1	KCl	lgK:2.976(4SF)	5	MGL	931
6	25	0.1	KCl	dH:-9.4(2SF)	5	ENS	931

Reaction No. 34723

H+1(-1)_B(OH)3 + 2<DFructose> = B+3_H+1(-4)_DFructose(2) + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:4.98(3SF)	3	MGL	140

Reaction No. 34725

B(OH)4-1 + 2<DGalactose> = B+3_H+1(-4)_DGalactose(2) + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.3(2SF)	1	ELC	
2	25	0.1	KCl	lgK:2.39(3SF)	0	MGL	931

Reaction No. 34728

H+1(-1)_B(OH)3 + DGalactose = BO+1_H+1(-2)_DGalactose + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:2.10(3SF)	5	MGL	140

Reaction No. 34729

H+1(-1)_B(OH)3 + 2<DGalactose> = B+3_H+1(-4)_DGalactose(2) + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:2.47(3SF)	5	MGL	931

Reaction No. 34731

B(OH)4-1 + 2<DGlucose> = B+3_H+1(-4)_DGlucose(2) + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.1	KCl	lgK:2.894(4SF)	5	MGL	931
2	15	0.1	KCl	lgK:2.750(4SF)	5	MGL	931
3	25	0.1	KCl	lgK:2.86(3SF)	5	MGL	931
4	25	0.1	KCl	lgK:2.633(4SF)	5	MGL	931
5	35	0.1	KCl	lgK:2.560(4SF)	5	MGL	931
6	45	0.1	KCl	lgK:2.407(4SF)	5	MGL	931
7	25	0.1	KCl	dH:-4.6(2SF)	5	ENS	931

Reaction No. 34732

B(OH)4-1 + DGlucose = B+3_H+1(-2)_DGlucose_OH-1(2) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.1	KCl	lgK:2.305(4SF)	5	MGL	931
2	15	0.1	KCl	lgK:2.071(4SF)	5	MGL	931
3	25	0.1	KCl	lgK:2.022(4SF)	5	MGL	931

4	35	0.1	KCl	lgK:1.985(4SF)	5	MGL	931
5	45	0.1	KCl	lgK:1.978(4SF)	5	MGL	931
6	25	0.1	KCl	dH:-1.3(2SF)	5	ENS	931

Reaction No. 34735

H+1(-1)_B(OH)3 + DGlucose = BO+1_H+1(-2)_DGlucose + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:1.90(3SF)	5	MGL	140

Reaction No. 34736

H+1(-1)_B(OH)3 + 2<DGlucose> = B+3_H+1(-4)_DGlucose(2) + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.8(1SF)	1	ELC	
2	25	0.1	Unknown	lgK:2.89(3SF)	0	MGL	140

Reaction No. 34740

B(OH)4-1 + 2<DMannose> = B+3_H+1(-4)_DMannose(2) + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:4.52(3SF)	5	MGL	931

Reaction No. 34742

H+1(-1)_B(OH)3 + DMannose = BO+1_H+1(-2)_DMannose + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:1.70(3SF)	5	MGL	140

Reaction No. 34743

H+1(-1)_B(OH)3 + 2<DMannose> = B+3_H+1(-4)_DMannose(2) + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:2.69(3SF)	5	MGL	140

Reaction No. 37537

BO+1_OH-1(2) + H+1(2)_Cat-2 = BO+1_Cat-2 + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:3.89(3SF)	4	MGL	140

Reaction No. 37538

BO+1_OH-1(2) + 2<H+1(2)_Cat-2> = B+3_Cat-2(2) + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0.1	Unknown	lgK:4.15(3SF)	4	MGL	140

Reaction No. 37878

B(s) + F2(g) + H2(g) + O2(g) + e-1 = B(OH)2F2-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-318.56(5SF)	5	CRV	406

Reaction No. 37880

B(s) + 0.5<F2(g)> + 1.5<H2(g)> + 1.5<O2(g)> + e-1 = B(OH)3F-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:220.9(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-297.84(5SF)	0	CRV	406

Reaction No. 37881

B(OH)4-1 + 4<F-1> = BF4-1 + 4<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.5(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-26.636(5SF)	0	CRV	406
3	25	1	NaCl	lgK:-27.1(3SF)	5	EOD	406

Reaction No. 37882

B(OH)4-1 + 3<F-1> = B(OH)F3-1 + 3<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-18.(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-24.61(4SF)	0	CRV	406
Note (2): Self-inconsistent it seems							
3	25	1	NaCl	lgK:-18.70(4SF)	0	EOD	406

Reaction No. 39122

B(OH)3F-1 + F-1 = B(OH)2F2-1 + OH-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.58(3SF)	3	CRV	406
2	25	0	Inf. Dilution	lgK:-6.4(2SF)	0	EOD	406
Note (2): Suspect other species							

Reaction No. 39123

B(OH)4-1 + 2<F-1> = B(OH)2F2-1 + 2<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-12.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-15.19(4SF)	0	CRV	406
3	25	1	NaCl	lgK:-11.6(3SF)	0	EOD	406

Reaction No. 39641							
DGalactose + B(OH)4-1 = B(OH)4-1_DGalactose							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaNO3	lgK:2.21(0.03SD)	5	MPT	2198
2	25	0.1	NaNO3	dG:-12.6(0.2SD)kJ	5	EFE	2198
3	25	0.1	NaNO3	dH:-24.7(1.7SD)kJ	5	MCL	2198
4	25	0.1	NaNO3	dS:-42.(4.SD)J	5	EFE	2198

Reaction No. 39642							
B(OH)4-1_DGalactose + DGalactose = B(OH)4-1_DGalactose(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaNO3	lgK:0.18(0.17SD)	5	MPT	2198
2	25	0.1	NaNO3	dG:-1.0(0.8SD)kJ	5	EFE	2198
3	25	0.1	NaNO3	dH:48.5(4.SD)kJ	5	MCL	2198
4	25	0.1	NaNO3	dS:167.(12.SD)J	5	EFE	2198

Reaction No. 39643							
DGlucose + B(OH)4-1 = B(OH)4-1_DGlucose							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.025	Unknown	lgK:2.33(3SF)	4	MGL	2118
2	13.05	0.025	Unknown	lgK:2.24(3SF)	4	MGL	2118
3	25	0.025	Unknown	lgK:2.13(3SF)	4	MGL	2118
4	25	0.1	NaNO3	lgK:2.27(0.05SD)	5	MPT	2198
5	34.98	0.025	Unknown	lgK:2.10(3SF)	4	MGL	2118
6	25	0.1	NaNO3	dG:-12.9(0.3SD)kJ	5	EFE	2198
7	0:34.98	0.025	Unknown	dH:-3.52(3SF)	4	MTD	2118
8	25	0.1	NaNO3	dH:-17.0(1.SD)kJ	5	MCL	2198
9	25	0.1	NaNO3	dS:-12.5(4.SD)J	5	EFE	2198

Reaction No. 39644							
B(OH)4-1_DGlucose + DGlucose = B(OH)4-1_DGlucose(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0.1	NaNO3	lgK:0.49(0.05SD)	5	MPT	2198
2	25	0.1	NaNO3	dG:-2.8(0.3SD)kJ	5	EFE	2198
3	25	0.1	NaNO3	dH:15.(3.SD)kJ	5	MCL	2198
4	25	0.1	NaNO3	dS:58.(8.SD)J	5	EFE	2198

Reaction No. 39645

DFructose + B(OH)4-1 = B(OH)4-1_DFructose

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaNO3	lgK:3.58(0.05SD)	5	MPT	2198
2	25	0.1	NaNO3	dG:-20.4(0.3SD)kJ	5	EFE	2198
3	25	0.1	NaNO3	dH:-3.(1.SD)kJ	5	MCL	2198
4	25	0.1	NaNO3	dS:59.(4.SD)J	5	EFE	2198

Reaction No. 39646

B(OH)4-1_DFructose + DFructose = B(OH)4-1_DFructose(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaNO3	lgK:1.36(0.05SD)	5	MPT	2198
2	25	0.1	NaNO3	dG:-7.7(0.3SD)kJ	5	EFE	2198
3	25	0.1	NaNO3	dH:-33.(2.SD)kJ	5	MCL	2198
4	25	0.1	NaNO3	dS:-84.(4.SD)J	5	EFE	2198

Reaction No. 39647

2<DMannose> + B(OH)4-1 = B(OH)4-1_DMannose(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaNO3	lgK:4.42(0.05SD)	5	MPT	2198
2	25	0.1	NaNO3	dG:-25.2(0.3SD)kJ	5	EFE	2198
3	25	0.1	NaNO3	dH:-6.81(0.02SD)kJ	5	MCL	2198
4	25	0.1	NaNO3	dS:61.5(0.8SD)J	5	EFE	2198

Reaction No. 39648

2<LSorbose> + B(OH)4-1 = B(OH)4-1_LSorbose(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaNO3	lgK:5.80(0.05SD)	5	MPT	2198
2	25	0.1	NaNO3	dG:-33.1(0.3SD)kJ	5	EFE	2198
3	25	0.1	NaNO3	dH:-25.15(0.08SD)kJ	5	MCL	2198

Reaction No. 40763

B(OH)4-1 + 2<LRhamnose> = B+3_H+1(-4)_LRhamnose(2) + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:2.61(3SF)	5	MGL	931

Reaction No. 40766

B(OH)4-1 + 2<LSorbose> = B+3_H+1(-4)_LSorbose(2) + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:5.80(3SF)	5	MGL	931

Reaction No. 42058

4<B(s)> + 8<H2(g)> + 8<O2(g)> + Zn(s) + 2<e-1> = Zn+2_B(OH)4-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1153.50(6SF)	5	CRV	406

Reaction No. 42059

4<B(s)> + Cd(s) + 8<H2(g)> + 8<O2(g)> + 2<e-1> = Cd+2_B(OH)4-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1135.30(6SF)	5	CRV	406

Reaction No. 42060

B(s) + Fe(s) + 2<H2(g)> + 2<O2(g)> - 2<e-1> = Fe+3_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:212.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-289.690(6SF)	0	CRV	406

Reaction No. 42061

2<B(s)> + Fe(s) + 4<H2(g)> + 4<O2(g)> - e-1 = Fe+3_B(OH)4-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:421.5(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-594.680(6SF)	0	CRV	406

Reaction No. 42876

Al(s) + 2<B(s)> + 4<H2(g)> + 4<O2(g)> - e-1 = Al+3_B(OH)4-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-687.910(6SF)	5	CRV	406

Reaction No. 43232

B(OH)3 + 2<H+1(2)_Salicylic-2> = B+3_Salicylic-2(2) + H+1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:0.7(1SF)	4	MGL	906

Reaction No. 43233

B(OH)3 + H+1(2)_Salicylic-2 = B+3_OH-1(2)_Salicylic-2 + H+1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:-1.62(3SF)	4	MGL	906

Reaction No. 43949

B(OH)3 + H+1(2)_4NitrCat-2 = B+3_OH-1(2)_4NitrCat-2 + H+1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5.3(2SF)	1	ELC	
2	25	0.1	KCl	lgK:-3.76(0.01SD)	0	MSP	906

Reaction No. 43950

B(OH)4-1 + H+1(2)_4NitrCat-2 = B+3_OH-1(2)_4NitrCat-2 + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:3.96(3SF)	4	MPL	906

Reaction No. 44174

B(OH)3 + H+1_Tartaric-2 = B+3_H+1_OH-1(3)_Tartaric-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:1.18(3SF)	4	MOR	906

Reaction No. 44175

B(OH)3 + Tartaric-2 = B+3_OH-1(3)_Tartaric-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:0.70(2SF)	4	MOR	906

Reaction No. 44592

Al(s) + B(s) + 2<H2(g)> + 2<O2(g)> - 2<e-1> = Al+3_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-402.850(6SF)	5	CRV	406

Reaction No. 44748

Al(s) + 3<B(s)> + 6<H2(g)> + 6<O2(g)> = Al+3_B(OH)4-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-970.740(6SF)	5	CRV	406

Reaction No. 45841

Al(s) + 6<B(s)> + 12<H2(g)> + 12<O2(g)> + 3<e-1> = Al+3_B(OH)4-1(6)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1822.50(6SF)	5	CRV	406

Reaction No. 45842

4<B(s)> + Co(s) + 8<H2(g)> + 8<O2(g)> + 2<e-1> = Co+2_B(OH)4-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1128.20(6SF)	5	CRV	406

Reaction No. 45843

3<B(s)> + 6<H2(g)> + Ni(s) + 6<O2(g)> + e-1 = Ni+2_B(OH)4-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-849.260(6SF)	5	CRV	406

Reaction No. 45844

3<B(s)> + Cu(s) + 6<H2(g)> + 6<O2(g)> + e-1 = Cu+2_B(OH)4-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-862.780(6SF)	0	CRV	406

Note (1): Far too high - see B110 & B120 for the story

Reaction No. 45845

2<B(s)> + Cu(s) + 4<H2(g)> + 4<O2(g)> = Cu+2_B(OH)4-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:400.(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-583.510(6SF)	0	CRV	406

Note (2): Low wgt for inter-reaction consistency - see B110 & B120

Reaction No. 45881

B(s) + Cu(s) + 2<H2(g)> + 2<O2(g)> - e-1 = Cu+2_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:195.(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-300.740(6SF)	0	CRV	406

Note (2): Low wgt for inter-reaction consistency; see B110 & B120

Reaction No. 45947

Ag(s) + B(s) + 2<H2(g)> + 2<O2(g)> = Ag+1_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:189.7(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-295.660(6SF)	0	CRV	406

Reaction No. 46360

3<B(s)> + 6<H2(g)> + 6<O2(g)> + Pb(s) + e-1 = Pb+2_B(OH)4-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-847.690(6SF)	5	CRV	406

Reaction No. 47032

B(s) + 2<H2(g)> + 2<O2(g)> + Pb(s) - e-1 = Pb+2_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:209.2(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-288.440(6SF)	0	CRV	406

Reaction No. 47628

B(s) + Ba(s) + 2<H2(g)> + 2<O2(g)> - e-1 = Ba+2_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-411.620(6SF)	5	CRV	406

Reaction No. 48197

B(s) + 2<H2(g)> + 2<O2(g)> + Sr(s) - e-1 = Sr+2_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:302.2(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-410.900(6SF)	0	CRV	406

Reaction No. 48513

B(s) + Ca(s) + 2<H2(g)> + 2<O2(g)> - e-1 = Ca+2_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-410.220(6SF)	5	CRV	406

Reaction No. 48787

B(s) + 2<H2(g)> + Mg(s) + 2<O2(g)> - e-1 = Mg+2_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:283.4(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-386.710(6SF)	0	CRV	406

Reaction No. 48819

B(s) + 2<H2(g)> + Na(s) + 2<O2(g)> = Na+1_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	Inert	lgK:247.0(4SF)	1	EOR	
2	100	0	Inf. Dilution	lgK:192.5(4SF)	1	EOR	
3	25	0	Inf. Dilution	dG:-338.580(6SF)	4	CRV	406

Reaction No. 48820

4<B(OH)3> + 2<OH-1> = B4O5(OH)4-2 + 5<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.94(4SF)	5	CRV	406
Note (1): Sign changed							
2	25	3	Na+1 salt	lgK:13.71(0.04MD)	6	MHY	22982

Reaction No. 48830

3<B(OH)3> + OH-1 = B3O3(OH)4-1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.5(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:6.69(3SF)	0	CRV	406
Note (2): Sign changed							

Reaction No. 49524

2<B(OH)3> + OH-1 = B2O(OH)5-1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.69(3SF)	5	CRV	406

Reaction No. 49533

H+1(-2)_B(OH)4-1 + H+1 = H+1(-1)_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.3(3SF)	0	MCO	8835
Note (1): Discredited by Mesmer et al [409]; Bassett [406] also							

Reaction No. 49534

H+1(-1)_B(OH)4-1 + H+1 = B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.3(3SF)	0	MCO	8835
Note (1): Discredited by Mesmer et al [409]; Bassett [406] also							

Reaction No. 52292							
2<H+1> + B(OH)4-1 + Cat-2 = H+1(2)_B(OH)4-1_Cat-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:27.46(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:26.47(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:26.32(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:26.24(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:26.18(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:26.17(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:26.16(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:26.18(0.01SD)	4	EDH	5390

Reaction No. 52293							
4<H+1> + B(OH)4-1 + 2<Cat-2> = H+1(4)_B(OH)4-1_Cat-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:51.38(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:49.37(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:49.07(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:48.94(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:48.84(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:48.85(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:48.87(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:48.90(0.01SD)	4	EDH	5390

Reaction No. 52294							
2<H+1> + B(OH)4-1 + Salicylic-2 = H+1(2)_B(OH)4-1_Salicylic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:24.25(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:23.66(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:23.54(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:23.48(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:23.44(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:23.42(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:23.40(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:23.41(0.01SD)	4	EDH	5390

Reaction No. 53506

H+1 + BF4-1 = H+1_BF4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.5(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:0.5(1SF)	4	CRV	6330
3	23		DMF	lgK:2.(1SF)	4	MSC	5398

Reaction No. 53507							
BF4-1 + Cu+1 = Cu+1_BF4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	MeCN Inf. Dilution	lgK:0.965(3SF)	4	MDG	5398

Reaction No. 53508							
BF4-1 + Tl+1 = Tl+1_BF4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	MeCN Inf. Dilution	lgK:1.15(3SF)	5	MCN	5398

Reaction No. 54528							
Mg+2 + B(OH)4-1 = Mg+2_B(OH)4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.618(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:1.34(3SF)	4	CRV	406
3	25	0	Inf. Dilution	lgK:1.52(3SF)	4	CRV	406
4	25	0	Inf. Dilution	lgK:1.63(3SF)	4	CRV	406
5	25	0	Inf. Dilution	lgK:1.57(0.02SD)	3	EES	7391
6	25	0	Inf. Dilution	lgK:1.3993(5SF)	3	EOD	30797
7	25	0	UCT_Sea	lgK:1.63(0.01SD)	4	EDH	5390
8	25	0.1	UCT_Sea	lgK:1.28(0.01SD)	4	EDH	5390
9	25	0.16	Cl-1 salt	lgK:1.6(0.05SD)w	0	MPH	373
10	25	0.2	UCT_Sea	lgK:1.22(0.01SD)	4	EDH	5390
11	25	0.3	UCT_Sea	lgK:1.19(0.01SD)	4	EDH	5390
12	25	0.4	UCT_Sea	lgK:1.16(0.01SD)	4	EDH	5390
13	25	0.5	UCT_Sea	lgK:1.15(0.01SD)	4	EDH	5390
14	25	0.6	UCT_Sea	lgK:1.14(0.01SD)	4	EDH	5390
15	25	0.7	NaCl	lgK:1.1(0.03SD)	3	MPS	769
16	25	0.7	Seawater	lgK:0.73(2SF)	0	ENS	405
17	25	0.7	UCT_Sea	lgK:1.13(0.01SD)	4	EDH	5390

18	25	0.7	Unknown	lgK:1.4(0.1SD)	0	CMN	406
19	25	0.7	Unknown	lgK:0.93(0.03SD)	5	CMN	7391
20	25	0.7	Unknown	lgK:0.73(2SF)	2	ENS	7391
21	25	0.7	Unknown	lgK:0.90(2SF)	3	ENS	7391
22	10:50	0.16	Cl-1 salt	dH:-0.5(1.SD)	5	MPH	373

Reaction No. 54529

Na+1 + B(OH)4-1 = Na+1_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.04	Unknown	lgK:1.87(3SF)	0	CRV	8836
2	25	0	Inf. Dilution	lgK:0.2368(4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:0.33(2SF)	4	CRV	406
4	25	0	Inf. Dilution	lgK:0.22(0.12SD)	4	CMN	7391
5	25	0	UCT_Sea	lgK:0.24(0.01SD)	4	EDH	5390
6	25	0.1	UCT_Sea	lgK:-0.02(0.01SD)	4	EDH	5390
7	25	0.2	UCT_Sea	lgK:-0.10(0.01SD)	4	EDH	5390
8	25	0.3	UCT_Sea	lgK:-0.17(0.01SD)	4	EDH	5390
9	25	0.4	UCT_Sea	lgK:-0.22(0.01SD)	4	EDH	5390
10	25	0.5	UCT_Sea	lgK:-0.27(0.01SD)	4	EDH	5390
11	25	0.6	UCT_Sea	lgK:-0.32(0.01SD)	4	EDH	5390
12	25	0.68	Unknown	lgK:-0.24(0.05SD)	4	ENS	407
13	25	0.7	UCT_Sea	lgK:-0.36(0.01SD)	4	EDH	5390
14	25	0.7	Unknown	lgK:-0.24(0.08SD)	6	CMN	7391
15	25	0.7	NaCl	lgR:-0.4(0.03SD)	3	MPS	769

Reaction No. 54530

Ca+2 + B(OH)4-1 = Ca+2_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.152(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:1.45(3SF)	3	CRV	406
3	25	0	Inf. Dilution	lgK:1.80(3SF)	4	CRV	406
4	25	0	Inf. Dilution	lgK:1.83(3SF)	4	CRV	406
5	25	0	Inf. Dilution	lgK:1.80(0.03SD)	4	CMN	7391
6	25	0	UCT_Sea	lgK:1.80(0.01SD)	4	EDH	5390
7	25	0.1	UCT_Sea	lgK:1.40(0.01SD)	4	EDH	5390
8	25	0.16	Cl-1 salt	lgK:1.8(0.05SD)w	5	MPH	373

9	25	0.2	UCT_Sea	lgK:1.29(0.01SD)	4	EDH	5390
10	25	0.3	UCT_Sea	lgK:1.23(0.01SD)	4	EDH	5390
11	25	0.4	UCT_Sea	lgK:1.18(0.01SD)	4	EDH	5390
12	25	0.5	UCT_Sea	lgK:1.13(0.01SD)	4	EDH	5390
13	25	0.6	UCT_Sea	lgK:1.09(0.01SD)	4	EDH	5390
14	25	0.7	NaCl	lgK:1.1(0.03SD)	5	MPS	769
15	25	0.7	UCT_Sea	lgK:1.06(0.01SD)	4	EDH	5390
16	25	0.7	Unknown	lgK:1.6(0.2SD)	5	CMN	406
17	25	0.7	Unknown	lgK:1.11(0.02SD)	6	CMN	7391
18	25	0.7	Unknown	lgK:0.41(2SF)	3	ENS	7391
19	25	0.7	Unknown	lgK:0.73(2SF)	3	ENS	7391
20	25	0.7	Unknown	lgK:1.11(3SF)	4	ENS	7391
21	10:50	0.16	Cl-1 salt	dH:-0.9(0.6SD)	5	MPH	373

Reaction No. 54531

Sr+2 + B(OH)4-1 = Sr+2_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.048(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:1.55(3SF)	4	CRV	406
3	25	0	Inf. Dilution	lgK:1.55(0.03SD)	4	CMN	7391
4	25	0	UCT_Sea	lgK:1.55(0.01SD)	4	EDH	5390
5	25	0.1	UCT_Sea	lgK:1.08(0.01SD)	4	EDH	5390
6	25	0.15	Cl-1 salt	lgK:1.6(0.05SD)w	5	MPH	373
7	25	0.2	UCT_Sea	lgK:0.94(0.01SD)	4	EDH	5390
8	25	0.3	UCT_Sea	lgK:0.84(0.01SD)	4	EDH	5390
9	25	0.4	UCT_Sea	lgK:0.75(0.01SD)	4	EDH	5390
10	25	0.5	UCT_Sea	lgK:0.61(0.01SD)	4	EDH	5390
11	25	0.6	UCT_Sea	lgK:0.61(0.01SD)	4	EDH	5390
12	25	0.7	NaCl	lgK:0.5(0.03SD)	5	MPS	769
13	25	0.7	UCT_Sea	lgK:0.54(0.01SD)	4	EDH	5390
14	25	0.7	Unknown	lgK:0.85(0.03SD)	3	EES	7391
15	10:50	0.15	Cl-1 salt	dH:-0.7(1.SD)	5	MPH	373

Reaction No. 54532

Ba+2 + B(OH)4-1 = Ba+2_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:1.933(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:1.49(3SF)	4	CRV	406
3	25	0	Inf. Dilution	lgK:1.49(0.03SD)	4	CMN	7391
4	25	0	UCT_Sea	lgK:1.49(0.01SD)	4	EDH	5390
5	25	0.1	UCT_Sea	lgK:1.02(0.01SD)	4	EDH	5390
6	25	0.15	Cl-1 salt	lgK:1.5(0.05SD)w	5	MPH	373
7	25	0.2	UCT_Sea	lgK:0.88(0.01SD)	4	EDH	5390
8	25	0.3	UCT_Sea	lgK:0.78(0.01SD)	4	EDH	5390
9	25	0.4	UCT_Sea	lgK:0.69(0.01SD)	4	EDH	5390
10	25	0.5	UCT_Sea	lgK:0.61(0.01SD)	4	EDH	5390
11	25	0.6	UCT_Sea	lgK:0.55(0.01SD)	4	EDH	5390
12	25	0.7	UCT_Sea	lgK:0.48(0.01SD)	4	EDH	5390
13	25	0.7	Unknown	lgK:0.73(0.03SD)	3	EES	7391
14	10:50	0.14	Cl-1 salt	dH:-0.7(1.SD)	5	MPH	373

Reaction No. 54533

Fe+3 + B(OH)3 + H2O = Fe+3_B(OH)4-1 + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01:0.2	Unknown	lgK:-0.7(0.2SD)	3	MSP	413
2	25	0.68	NaClO4	lgK:-1.97(0.1SD)	3	MSP	413
Note (2): Model queried by Byrne & Thompson, J. Soln. Chem., 1997, 26, 729 [7333]							
3	25	0.68	NaClO4	lgK:-2.24(3SF)	5	EOD	7333
Note (3): Recalculated from the data of Elrod, MSc., 1979; cf. [413]							
4	25	0.7	NaClO4/m	lgK:-2.268(4SF)	7	MPT	7333

Reaction No. 54534

Fe+3 + 2<B(OH)3> + 2<H2O> = Fe+3_B(OH)4-1(2) + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.8(2SF)	2	EDH	
2	25	0.68	NaClO4	lgK:-4.7(0.2SD)	3	MSP	413

Reaction No. 54535

B(OH)4-1 + Ag+1 = Ag+1_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.2(0.05SD)	4	MGL	408
2	25	0	Inf. Dilution	lgK:0.6(1SF)	3	ENS	6405
3	25	0	UCT_Sea	lgK:1.20(0.01SD)	4	EDH	5390

4	25	0.1	UCT_Sea	lgK:0.95(0.01SD)	4	EDH	5390
5	25	0.2	UCT_Sea	lgK:0.88(0.01SD)	4	EDH	5390
6	25	0.3	UCT_Sea	lgK:0.83(0.01SD)	4	EDH	5390
7	25	0.4	UCT_Sea	lgK:0.80(0.01SD)	4	EDH	5390
8	25	0.5	UCT_Sea	lgK:0.77(0.01SD)	4	EDH	5390
9	25	0.6	UCT_Sea	lgK:0.74(0.01SD)	4	EDH	5390
10	25	0.7	UCT_Sea	lgK:0.72(0.01SD)	4	EDH	5390
11	25	3	NaClO4	lgK:0.4(0.05SD)	5	MGL	408

Reaction No. 54537

Pb+2 + B(OH)4-1 = Pb+2_B(OH)4-1

Note: Turner & Vukadin (Mar. Chem., 1983, 14, 133) conclude that "if a complex PbB(OH)4+ exists, the log of the stability constant is less than 3.5 at an ionic strength of 0.7 M"

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.799(4SF)	1	EOR	
2	25	0	UCT_Sea	lgK:2.77(0.01SD)	3	EDH	5390
3	25	0.1	UCT_Sea	lgK:2.39(0.01SD)	3	EDH	5390
4	25	0.2	UCT_Sea	lgK:2.31(0.01SD)	3	EDH	5390
5	25	0.3	UCT_Sea	lgK:2.27(0.01SD)	3	EDH	5390
6	25	0.4	UCT_Sea	lgK:2.24(0.01SD)	3	EDH	5390
7	25	0.5	UCT_Sea	lgK:2.22(0.01SD)	3	EDH	5390
8	25	0.6	UCT_Sea	lgK:2.21(0.01SD)	3	EDH	5390
9	25	0.7	KNO3	lgK:2.2(0.2SD)	3	MSV	429
10	25	0.7	UCT_Sea	lgK:2.20(0.01SD)	3	EDH	5390

Reaction No. 54538

Pb+2 + 2<B(OH)4-1> = Pb+2_B(OH)4-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.344(4SF)	1	EOR	
2	25	0	UCT_Sea	lgK:5.32(0.01SD)	4	EDH	5390
3	25	0.1	UCT_Sea	lgK:4.72(0.01SD)	4	EDH	5390
4	25	0.2	UCT_Sea	lgK:4.59(0.01SD)	4	EDH	5390
5	25	0.3	UCT_Sea	lgK:4.52(0.01SD)	4	EDH	5390
6	25	0.4	UCT_Sea	lgK:4.47(0.01SD)	4	EDH	5390
7	25	0.5	UCT_Sea	lgK:4.44(0.01SD)	4	EDH	5390
8	25	0.6	UCT_Sea	lgK:4.42(0.01SD)	4	EDH	5390

9	25	0.7	KNO3	lgK:4.4(0.2SD)	3	MSV	429
10	25	0.7	UCT_Sea	lgK:4.41(0.01SD)	4	EDH	5390

Reaction No. 54539

Cu+2 + B(OH)4-1 = Cu+2_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.037(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:7.13(3SF)	0	EOD	406
Note (2): Data from Shchigol, RJIC, 65, 10, 1142 (might refer to B120)							
3	25	0	UCT_Sea	lgK:4.02(0.01SD)	4	EDH	5390
4	25	0.1	UCT_Sea	lgK:3.64(0.01SD)	4	EDH	5390
5	25	0.2	UCT_Sea	lgK:3.56(0.01SD)	4	EDH	5390
6	25	0.3	UCT_Sea	lgK:3.52(0.01SD)	4	EDH	5390
7	25	0.4	UCT_Sea	lgK:3.49(0.01SD)	4	EDH	5390
8	25	0.5	UCT_Sea	lgK:3.47(0.01SD)	4	EDH	5390
9	25	0.6	UCT_Sea	lgK:3.46(0.01SD)	4	EDH	5390
10	25	0.7	KNO3	lgK:3.5(0.2SD)	3	MSV	429
11	25	0.7	UCT_Sea	lgK:3.43(0.01SD)	4	EDH	5390
12	25	0.7	Unknown	lgK:3.48(3SF)	5	CRV	7271

Reaction No. 54540

Cu+2 + 2<B(OH)4-1> = Cu+2_B(OH)4-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.068(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:12.45(4SF)	0	EOD	406
Note (2): Data from Shchigol, RJIC, 65, 10, 1142 (might refer to B130)							
3	25	0	UCT_Sea	lgK:7.04(0.01SD)	4	EDH	5390
4	25	0.1	UCT_Sea	lgK:6.44(0.01SD)	4	EDH	5390
5	25	0.2	UCT_Sea	lgK:6.31(0.01SD)	4	EDH	5390
6	25	0.3	UCT_Sea	lgK:6.24(0.01SD)	4	EDH	5390
7	25	0.4	UCT_Sea	lgK:6.19(0.01SD)	4	EDH	5390
8	25	0.5	UCT_Sea	lgK:6.16(0.01SD)	4	EDH	5390
9	25	0.6	UCT_Sea	lgK:6.14(0.01SD)	4	EDH	5390
10	25	0.7	KNO3	lgK:6.1(0.2SD)	3	MSV	429
11	25	0.7	UCT_Sea	lgK:6.13(0.01SD)	4	EDH	5390
12	25	0.7	Unknown	lgK:6.13(3SF)	5	CRV	7271

Reaction No. 54541							
Cd+2 + B(OH)4-1 = Cd+2_B(OH)4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.024(4SF)	1	EOR	
2	25	0	UCT_Sea	lgK:1.99(0.01SD)	4	EDH	5390
3	25	0.1	UCT_Sea	lgK:1.61(0.01SD)	4	EDH	5390
4	25	0.2	UCT_Sea	lgK:1.53(0.01SD)	4	EDH	5390
5	25	0.3	UCT_Sea	lgK:1.49(0.01SD)	4	EDH	5390
6	25	0.4	UCT_Sea	lgK:1.46(0.01SD)	4	EDH	5390
7	25	0.5	UCT_Sea	lgK:1.44(0.01SD)	4	EDH	5390
8	25	0.6	UCT_Sea	lgK:1.43(0.01SD)	4	EDH	5390
9	25	0.7	KNO3	lgK:1.4(0.2SD)	3	MSV	429
10	25	0.7	UCT_Sea	lgK:1.42(0.01SD)	4	EDH	5390

Reaction No. 54542							
Cd+2 + 2<B(OH)4-1> = Cd+2_B(OH)4-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.631(4SF)	1	EOR	
2	25	0	UCT_Sea	lgK:3.61(0.01SD)	4	EDH	5390
3	25	0.1	UCT_Sea	lgK:3.01(0.01SD)	4	EDH	5390
4	25	0.2	UCT_Sea	lgK:2.88(0.01SD)	4	EDH	5390
5	25	0.3	UCT_Sea	lgK:2.81(0.01SD)	4	EDH	5390
6	25	0.4	UCT_Sea	lgK:2.76(0.01SD)	4	EDH	5390
7	25	0.5	UCT_Sea	lgK:2.73(0.01SD)	4	EDH	5390
8	25	0.6	UCT_Sea	lgK:2.71(0.01SD)	4	EDH	5390
9	25	0.7	KNO3	lgK:2.7(0.2SD)	3	MSV	429
10	25	0.7	UCT_Sea	lgK:2.70(0.01SD)	4	EDH	5390

Reaction No. 54543							
Zn+2 + B(OH)4-1 = Zn+2_B(OH)4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.499(4SF)	1	EOR	
2	25	0	UCT_Sea	lgK:1.47(0.01SD)	4	EDH	5390
3	25	0.1	UCT_Sea	lgK:1.09(0.01SD)	4	EDH	5390
4	25	0.2	UCT_Sea	lgK:1.01(0.01SD)	4	EDH	5390
5	25	0.3	UCT_Sea	lgK:0.97(0.01SD)	4	EDH	5390

6	25	0.4	UCT_Sea	lgK:0.94(0.01SD)	4	EDH	5390
7	25	0.5	UCT_Sea	lgK:0.92(0.01SD)	4	EDH	5390
8	25	0.6	UCT_Sea	lgK:0.91(0.01SD)	4	EDH	5390
9	25	0.7	KNO3	lgK:0.9(0.2SD)	3	MSV	429
10	25	0.7	UCT_Sea	lgK:0.90(0.01SD)	4	EDH	5390

Reaction No. 54544

Zn+2 + 2<B(OH)4-1> = Zn+2_B(OH)4-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.256(4SF)	1	EOR	
2	25	0	UCT_Sea	lgK:4.23(0.01SD)	4	EDH	5390
3	25	0.1	UCT_Sea	lgK:3.63(0.01SD)	4	EDH	5390
4	25	0.2	UCT_Sea	lgK:3.50(0.01SD)	4	EDH	5390
5	25	0.3	UCT_Sea	lgK:3.43(0.01SD)	4	EDH	5390
6	25	0.4	UCT_Sea	lgK:3.38(0.01SD)	4	EDH	5390
7	25	0.5	UCT_Sea	lgK:3.35(0.01SD)	4	EDH	5390
8	25	0.6	UCT_Sea	lgK:3.33(0.01SD)	4	EDH	5390
9	25	0.7	KNO3	lgK:3.3(0.2SD)	3	MSV	429
10	25	0.7	UCT_Sea	lgK:3.32(0.01SD)	4	EDH	5390

Reaction No. 54545

F-1 + B(OH)3 = B(OH)3_F-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.36(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:1.341(4SF)	0	CRV	406
Note (2): Low wgt for internal consistency							
3	25	0	Inf. Dilution	lgK:-0.36(2SF)	5	CRV	406
4	25	0	CHEMVAL4	lgK:-0.36(0.01SD)	3	ENS	5156
5	25	1	NaCl	lgK:-0.4(0.2SD)	6	MPT	5382
6	25	1	NaNO3	lgK:-0.3(0.06SD)	5	MMS	406

Reaction No. 54546

B(OH)3 + 2<F-1> = B(OH)2F2-1 + OH-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.7(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-8.24(3SF)	0	CRV	406
3	25	1	NaNO3	lgK:-6.27(0.05SD)	2	MIS	406

Reaction No. 54547

B(OH)3 + 3<F-1> = B(OH)F3-1 + 2<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.8(3SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:-13.08(4SF)	0	CRV	406
3	25	1	NaNO3	lgK:-14.23(0.05SD)	3	EOD	406

Reaction No. 54548

B(OH)3 + 4<F-1> = BF4-1 + 3<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-23.9(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-21.0(3SF)	0	CRV	406
3	25	1	NaNO3	lgK:-21.62(0.05SD)	3	EOD	406

Reaction No. 54607

Li+1 + B(OH)4-1 = Li+1_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	P=57.6 Inf. Dilution	lgK:0.338(0.197SD)	4	MCN	29937
2	25.1	0	P=195.2 Inf. Dilution	lgK:0.364(0.131SD)	4	MCN	29937
3	175	0	P=201.1 Inf. Dilution	lgK:0.57(0.17SD)	4	MCN	29937
4	250	0	P=203.1 Inf. Dilution	lgK:0.70(0.23SD)	4	MCN	29937
5	275.1	0	P=203.1 Inf. Dilution	lgK:1.23(0.08SD)	4	MCN	29937
6	5	0	Inf. Dilution	lgK:1.036(4SF)	4	MPK	9162
7	15	0	Inf. Dilution	lgK:1.038(4SF)	4	MPK	9162
8	25	0	Inf. Dilution	lgK:1.026(4SF)	1	EOR	
9	25	0	Inf. Dilution	lgK:1.043(4SF)	4	MPK	9162
10	25	0	UCT_Sea	lgK:0.55(0.01SD)	0	EDH	5390
11	25	0.1	UCT_Sea	lgK:0.29(0.01SD)	0	EDH	5390
12	25	0.2	UCT_Sea	lgK:0.21(0.01SD)	0	EDH	5390
13	25	0.3	UCT_Sea	lgK:0.14(0.01SD)	0	EDH	5390
14	25	0.4	UCT_Sea	lgK:0.09(0.01SD)	3	EDH	5390
15	25	0.5	UCT_Sea	lgK:0.04(0.01SD)	3	EDH	5390
16	25	0.6	UCT_Sea	lgK:-0.01(0.01SD)	3	EDH	5390

17	25	0.7	NaCl	lgK:-0.05(0.03SD)	5	MPS	769
18	25	0.7	UCT_Sea	lgK:-0.05(0.01SD)	3	EDH	5390
19	35	0	Inf. Dilution	lgK:1.046(4SF)	4	MPK	9162
20	45	0	Inf. Dilution	lgK:1.053(4SF)	4	MPK	9162
21	5	0	Inf. Dilution	dH:0.26(2SF)kJ	4	MTD	9162
22	15	0	Inf. Dilution	dH:0.49(2SF)kJ	4	MTD	9162
23	25	0	Inf. Dilution	dH:0.73(2SF)kJ	4	MTD	9162
24	35	0	Inf. Dilution	dH:0.97(2SF)kJ	4	MTD	9162
25	45	0	Inf. Dilution	dH:1.22(3SF)kJ	4	MTD	9162

Reaction No. 54624

B(s) + 2<H2(g)> + 2<O2(g)> + e-1 = B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-275.65(5SF)	5	CRV	406
2	25	0	Inf. Dilution	dG:-275.7(0.1SD)	4	ENS	3006
3	25	0	Inf. Dilution	dH:-321.2(0.1SD)	4	ENS	3006

Reaction No. 54647

BF4-1 + 3<e-1> = B(s) + 4<F-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-63.5(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-64.8(3SF)	0	EOR	
3	25	0	Inf. Dilution	E:-1.04(0.01SD)	0	MDG	775
4	25	0	Inf. Dilution	E:-1.284(4SF)	0	CRV	22551

Reaction No. 54965

Fe+3 + 3<B(OH)3> + 3<H2O> = Fe+3_B(OH)4-1(3) + 3<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01:0.2	Unknown	lgK:-7.(0.5SD)	0	MSP	413

Reaction No. 55739

10<B(s)> + 7<H2(g)> = B10H14(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:45.9(0.1SD)	4	ENS	3006
2	25	0	Inf. Dilution	dH:-10.8(0.1SD)	4	ENS	3006

Reaction No. 56038

B(OH)3 + 2<F-1> + H+1 = B(OH)2F2-1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL4	lgK:7.33(0.01SD)	4	ENS	5156
2	25	1	NaCl	lgK:7.06(0.02SD)	6	MPT	5382

Reaction No. 56039

B(OH)3 + 3<F-1> + 2<H+1> = B(OH)F3-1 + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL4	lgK:14.20(0.01SD)	4	ENS	5156
2	25	1	NaCl	lgK:13.69(0.003SD)	6	MPT	5382

Reaction No. 56040

B(OH)3 + 4<F-1> + 3<H+1> = BF4-1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	0	Inf. Dilution	lgK:19.4(3SF)	0	MHY	140
2	25	0	Inf. Dilution	lgK:18.4(3SF)	1	ELC	
3	25	0	CHEMVAL4	lgK:19.80(0.01SD)	0	ENS	5156
4	25	1	NaCl	lgK:19.0(0.1SD)	6	MPT	5382

Reaction No. 57514

H+1_ClO-1 + B(OH)3 = B(OH)3_ClO-1 + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaCl	lgK:2.25(0.01SD)	6	MSP	5289

Reaction No. 57515

B(OH)3 + H+1_BrO-1 = B(OH)3_BrO-1 + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaCl	lgK:1.8(0.04SD)	6	MSP	5289

Reaction No. 57730

B(OH)4-1 + F-1 = B(OH)3_F-1 + OH-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5.025(4SF)	5	CRV	406
2	25	1	NaCl	lgK:-5.3(0.2SD)	6	MPT	5382

Reaction No. 57731

B(OH)3_F-1 + F-1 = B(OH)2F2-1 + OH-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaCl	lgK:-6.3(0.2SD)	6	MPT	5382

Reaction No. 57732

B(OH)2F2-1 + F-1 = B(OH)F3-1 + OH-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaCl	lgK:-7.10(0.02SD)	6	MPT	5382

Reaction No. 57733

B(OH)F3-1 + F-1 = BF4-1 + OH-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.9(2SF)	1	ELC	
2	25	1	NaCl	lgK:-8.4(0.1SD)	4	MPT	5382

Reaction No. 57983

B(s) = B(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:521.0(4SF)kJ	3	CRV	8746
2	25	0	Inf. Dilution	dG:521.01(5.000SD)kJ	6	CRV	14923
3	25	0	Inf. Dilution	dH:130.(3.SD)	0	CRV	5605
4	25	0	Inf. Dilution	dH:565.(5.MD)kJ	3	CRV	8746
5	25	0	Inf. Dilution	dH:565.00(5.000SD)kJ	6	CRV	14923
6	25	0	Inf. Dilution	Cp:20.796(0.003SD)J/K	5	CRV	27292

Reaction No. 57984

2<B(s)> + 1.5<O2(g)> = B2O3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:209.0(4SF)	3	ENS	6422
2	25	0	Inf. Dilution	lgK:209.2(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:207.6(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:207.9(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:152.2(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:152.4(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:119.0(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:119.2(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:96.87(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:97.01(4SF)	0	ENS	6494

11	25	0	Inf. Dilution	dG:-1194.3(5SF)kJ	3	CRV	6330
12	25	0	Inf. Dilution	dG:-285.1(4SF)	2	EFE	6422
13	25	0	Inf. Dilution	dG:-285.5(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dG:-1193.70(6SF)kJ	4	ENS	7381
15	25	0	Inf. Dilution	dG:-1194.(4SF)kJ	6	CRV	8746
16	25	0	Inf. Dilution	dG:-1195.3(1.404SD)kJ	3	CRV	14923
17	25	0	Inf. Dilution	dG:-1194.30(0.7SD)kJ	5	CRV	27292
18	25	0	Inf. Dilution	dG:-1194.3(5SF)kJ	5	EOD	29520
19	25	0	Inf. Dilution	dH:-304.(0.3SD)	3	CRV	5605
20	25	0	Inf. Dilution	dH:-1273.5(5SF)kJ	3	CRV	6330
21	25	0	Inf. Dilution	dH:-304.0(4SF)	4	MTD	6422
22	25	0	Inf. Dilution	dH:-304.4(4SF)	4	MTD	6494
23	25	0	Inf. Dilution	dH:-1272.80(6SF)kJ	4	ENS	7381
24	25	0	Inf. Dilution	dH:-1273.5(1.4MD)kJ	6	CRV	8746
25	25	0	Inf. Dilution	dH:-1273.5(1.400SD)kJ	6	CRV	14923
26	25	0	Inf. Dilution	dH:-1273.50(0.7SD)kJ	5	CRV	27292
27	25	0	Inf. Dilution	dH:-1273.5(5SF)kJ	5	EOD	29520
28	127	0	Inf. Dilution	dH:-304.0(4SF)	1	MTD	6422
29	127	0	Inf. Dilution	dH:-304.4(4SF)	1	MTD	6494
30	227	0	Inf. Dilution	dH:-303.9(4SF)	0	MTD	6422
31	227	0	Inf. Dilution	dH:-304.4(4SF)	0	MTD	6494
32	327	0	Inf. Dilution	dH:-303.8(4SF)	0	MTD	6422
33	327	0	Inf. Dilution	dH:-304.2(4SF)	0	MTD	6494
34	25	0	Inf. Dilution	Cp:62.761(0.15SD)J/K	5	CRV	27292

Reaction No. 57985

B(s) + 1.5<F2(g)> = BF3(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1119.(4SF)kJ	6	CRV	8746
2	25	0	Inf. Dilution	dG:-1120.40(6SF)kJ	3	CRV	10802
3	25	0	Inf. Dilution	dG:-1119.4(0.803SD)kJ	6	CRV	14923
4	25	0	Inf. Dilution	dH:-271.5(0.2SD)	7	CRV	5605
5	25	0	Inf. Dilution	dH:-1123.(4SF)kJ	0	EFE	7276
6	25	0	Inf. Dilution	dH:-1136.0(0.8MD)kJ	6	CRV	8746
7	25	0	Inf. Dilution	dH:-1137.00(6SF)kJ	3	CRV	10802
8	25	0	Inf. Dilution	dH:-1136.0(0.800SD)kJ	6	CRV	14923

9	25	0	Inf. Dilution	Cp:50.463(0.05SD)J/K	5	CRV	27292

Reaction No. 58063

H+1 + F-1 + B(OH)4-1 = H+1_B(OH)4-1_F-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.826(4SF)	1	EOR	
2	25	0	UCT_Sea	lgK:8.86(0.01SD)	4	EDH	5390
3	25	0.1	UCT_Sea	lgK:8.59(0.01SD)	4	EDH	5390
4	25	0.2	UCT_Sea	lgK:8.56(0.01SD)	4	EDH	5390
5	25	0.3	UCT_Sea	lgK:8.54(0.01SD)	4	EDH	5390
6	25	0.4	UCT_Sea	lgK:8.52(0.01SD)	4	EDH	5390
7	25	0.5	UCT_Sea	lgK:8.51(0.01SD)	4	EDH	5390
8	25	0.6	UCT_Sea	lgK:8.51(0.01SD)	4	EDH	5390
9	25	0.7	UCT_Sea	lgK:8.52(0.01SD)	4	EDH	5390

Reaction No. 58064

2<H+1> + 2<F-1> + B(OH)4-1 = H+1(2)_B(OH)4-1_F-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.6(4SF)	1	EOR	
2	25	0	UCT_Sea	lgK:16.62(0.01SD)	4	EDH	5390
3	25	0.1	UCT_Sea	lgK:16.18(0.01SD)	4	EDH	5390
4	25	0.2	UCT_Sea	lgK:16.13(0.01SD)	4	EDH	5390
5	25	0.3	UCT_Sea	lgK:16.09(0.01SD)	4	EDH	5390
6	25	0.4	UCT_Sea	lgK:16.06(0.01SD)	4	EDH	5390
7	25	0.5	UCT_Sea	lgK:16.07(0.01SD)	4	EDH	5390
8	25	0.6	UCT_Sea	lgK:16.07(0.01SD)	4	EDH	5390
9	25	0.7	UCT_Sea	lgK:16.09(0.01SD)	4	EDH	5390

Reaction No. 58065

3<H+1> + 3<F-1> + B(OH)4-1 = H+1(3)_B(OH)4-1_F-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.97(4SF)	1	EOR	
2	25	0	UCT_Sea	lgK:22.99(0.01SD)	4	EDH	5390
3	25	0.1	UCT_Sea	lgK:22.36(0.01SD)	4	EDH	5390
4	25	0.2	UCT_Sea	lgK:22.27(0.01SD)	4	EDH	5390
5	25	0.3	UCT_Sea	lgK:22.23(0.01SD)	4	EDH	5390
6	25	0.4	UCT_Sea	lgK:22.21(0.01SD)	4	EDH	5390

7	25	0.5	UCT_Sea	lgK:22.22(0.01SD)	4	EDH	5390
8	25	0.6	UCT_Sea	lgK:22.23(0.01SD)	4	EDH	5390
9	25	0.7	UCT_Sea	lgK:22.25(0.01SD)	4	EDH	5390

Reaction No. 58066

4<H+1> + 4<F-1> + B(OH)4-1 = H+1(4)_B(OH)4-1_F-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.95(4SF)	1	EOR	
2	25	0	UCT_Sea	lgK:28.95(0.01SD)	4	EDH	5390
3	25	0.1	UCT_Sea	lgK:28.15(0.01SD)	4	EDH	5390
4	25	0.2	UCT_Sea	lgK:28.03(0.01SD)	4	EDH	5390
5	25	0.3	UCT_Sea	lgK:27.98(0.01SD)	4	EDH	5390
6	25	0.4	UCT_Sea	lgK:27.96(0.01SD)	4	EDH	5390
7	25	0.5	UCT_Sea	lgK:27.97(0.01SD)	4	EDH	5390
8	25	0.6	UCT_Sea	lgK:27.98(0.01SD)	4	EDH	5390
9	25	0.7	UCT_Sea	lgK:28.02(0.01SD)	4	EDH	5390

Reaction No. 58115

Fe+3 + B(OH)4-1 = Fe+3_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.541(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:7.531(4SF)	4	EDH	7333
3	25	0	UCT_Sea	lgK:7.44(0.01SD)	3	EDH	5390
4	25	0.1	NaNO3	lgK:3.67(3SF)	0	MGL	22560
5	25	0.1	UCT_Sea	lgK:6.99(0.01SD)	3	EDH	5390
6	25	0.2	UCT_Sea	lgK:6.92(0.01SD)	3	EDH	5390
7	25	0.3	UCT_Sea	lgK:6.89(0.01SD)	3	EDH	5390
8	25	0.4	UCT_Sea	lgK:6.87(0.01SD)	3	EDH	5390
9	25	0.5	UCT_Sea	lgK:6.86(0.01SD)	3	EDH	5390
10	25	0.6	UCT_Sea	lgK:6.85(0.01SD)	3	EDH	5390
11	25	0.7	Seawater/m	lgK:6.591(4SF)	0	CRV	7333
12	25	0.7	UCT_Sea	lgK:6.85(0.01SD)	3	EDH	5390
13	25	0.7	Unknown	lgK:10.85(4SF)	0	CRV	2556

Reaction No. 58116

Fe+3 + 2<B(OH)4-1> = Fe+3_B(OH)4-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:14.7(3SF)	1	ELC	
2	25	0	UCT_Sea	lgK:14.17(0.01SD)	0	EDH	5390
3	25	0.1	UCT_Sea	lgK:13.32(0.01SD)	0	EDH	5390
4	25	0.2	UCT_Sea	lgK:13.18(0.01SD)	0	EDH	5390
5	25	0.3	UCT_Sea	lgK:13.11(0.01SD)	0	EDH	5390
6	25	0.4	UCT_Sea	lgK:13.06(0.01SD)	0	EDH	5390
7	25	0.5	UCT_Sea	lgK:13.03(0.01SD)	0	EDH	5390
8	25	0.6	UCT_Sea	lgK:13.01(0.01SD)	0	EDH	5390
9	25	0.7	UCT_Sea	lgK:13.00(0.01SD)	0	EDH	5390
10	25	0.7	Unknown	lgK:22.4(3SF)	0	CRV	2556

Reaction No. 58122

H+1(2)_B(OH)4-1(3) + B(OH)4-1 = H+1(2)_B(OH)4-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.51(3SF)	3	CRV	552
2	25	1	KCl	lgK:0.91(2SF)	5	CRV	552
3	25	3	NaClO4	lgK:0.69(2SF)	5	CRV	552
4	25	1	KCl	dH:2.(1SF)	5	CRV	552

Reaction No. 58123

H+1 + 2<B(OH)4-1> = H+1_B(OH)4-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.675(4SF)	1	ELC	
2	25	0	UCT_Sea	lgK:9.12(0.01SD)	0	EDH	5390
3	25	0.1	UCT_Sea	lgK:8.84(0.01SD)	0	EDH	5390
4	25	0.2	UCT_Sea	lgK:8.80(0.01SD)	0	EDH	5390
5	25	0.3	UCT_Sea	lgK:8.77(0.01SD)	0	EDH	5390
6	25	0.4	UCT_Sea	lgK:8.73(0.01SD)	0	EDH	5390
7	25	0.5	UCT_Sea	lgK:8.71(0.01SD)	0	EDH	5390
8	25	0.6	UCT_Sea	lgK:8.70(0.01SD)	0	EDH	5390
9	25	0.7	UCT_Sea	lgK:8.69(0.01SD)	0	EDH	5390

Reaction No. 58124

2<H+1> + 3<B(OH)4-1> = H+1(2)_B(OH)4-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.32(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:20.4(3SF)	4	ENS	6405

3	25	0	UCT_Sea	lgK:20.69(0.01SD)	0	EDH	5390
4	25	0.1	UCT_Sea	lgK:20.14(0.01SD)	0	EDH	5390
5	25	0.2	UCT_Sea	lgK:20.05(0.01SD)	0	EDH	5390
6	25	0.3	UCT_Sea	lgK:19.99(0.01SD)	0	EDH	5390
7	25	0.4	UCT_Sea	lgK:19.92(0.01SD)	0	EDH	5390
8	25	0.5	UCT_Sea	lgK:19.88(0.01SD)	0	EDH	5390
9	25	0.6	UCT_Sea	lgK:19.86(0.01SD)	0	EDH	5390
10	25	0.7	UCT_Sea	lgK:19.85(0.01SD)	0	EDH	5390
11	25	3	Inert	lgK:21.6(3SF)	1	EOR	

Reaction No. 58125

2<H+1> + 4<B(OH)4-1> = H+1(2)_B(OH)4-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.0(3SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:21.0(3SF)	3	ENS	6405
3	25	0	UCT_Sea	lgK:20.66(0.01SD)	0	EDH	5390
4	25	0.1	UCT_Sea	lgK:20.32(0.01SD)	0	EDH	5390
5	25	0.2	UCT_Sea	lgK:20.31(0.01SD)	0	EDH	5390
6	25	0.3	UCT_Sea	lgK:20.30(0.01SD)	0	EDH	5390
7	25	0.4	UCT_Sea	lgK:20.30(0.01SD)	0	EDH	5390
8	25	0.5	UCT_Sea	lgK:20.31(0.01SD)	0	EDH	5390
9	25	0.6	UCT_Sea	lgK:20.31(0.01SD)	0	EDH	5390
10	25	0.7	UCT_Sea	lgK:20.32(0.01SD)	0	EDH	5390
11	25	3	Inert	lgK:20.6(3SF)	1	EOR	

Reaction No. 59167

EtGlycol + B(OH)4-1 = B(OH)4-1_EtGlycol

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.025	Unknown	lgK:0.52(2SF)	4	MGL	2118
2	13.05	0.025	Unknown	lgK:0.46(2SF)	4	MGL	2118
3	25	0.025	Unknown	lgK:0.33(2SF)	4	MGL	2118
4	34.98	0.025	Unknown	lgK:0.27(2SF)	4	MGL	2118
5	0:34.98	0.025	Unknown	dH:-2.7(2SF)	4	MTD	2118

Reaction No. 59168

2<EtGlycol> + B(OH)4-1 = B(OH)4-1_EtGlycol(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	0	0.025	Unknown	lgK:0.14(2SF)	4	MGL	2118
2	13.05	0.025	Unknown	lgK:0.08(1SF)	4	MGL	2118
3	25	0.025	Unknown	lgK:0.06(1SF)	4	MGL	2118
4	34.98	0.025	Unknown	lgK:-0.05(1SF)	4	MGL	2118
5	0:34.98	0.025	Unknown	dH:-1.97(3SF)	4	MTD	2118

Reaction No. 59169

12PrDiol + B(OH)4-1 = 12PrDiol_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.025	Unknown	lgK:0.80(2SF)	4	MGL	2118
2	13.05	0.025	Unknown	lgK:0.64(2SF)	4	MGL	2118
3	25	0.025	Unknown	lgK:0.61(2SF)	4	MGL	2118
4	34.98	0.025	Unknown	lgK:0.53(2SF)	4	MGL	2118
5	0:34.98	0.025	Unknown	dH:-3.02(3SF)	4	MTD	2118

Reaction No. 59170

2<12PrDiol> + B(OH)4-1 = 12PrDiol(2)_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.025	Unknown	lgK:0.92(2SF)	4	MGL	2118
2	13.05	0.025	Unknown	lgK:0.78(2SF)	4	MGL	2118
3	25	0.025	Unknown	lgK:0.59(2SF)	4	MGL	2118
4	34.98	0.025	Unknown	lgK:0.37(2SF)	4	MGL	2118
5	0:34.98	0.025	Unknown	dH:-7.2(2SF)	4	MTD	2118

Reaction No. 59171

13PrDiol + B(OH)4-1 = 13PrDiol_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.025	Unknown	lgK:0.45(2SF)	4	MGL	2118
2	13.05	0.025	Unknown	lgK:0.25(2SF)	4	MGL	2118
3	25	0.025	Unknown	lgK:0.10(2SF)	4	MGL	2118
4	34.98	0.025	Unknown	lgK:-0.02(1SF)	4	MGL	2118
5	0:34.98	0.025	Unknown	dH:-4.55(3SF)	4	MTD	2118

Reaction No. 59172

2<13PrDiol> + B(OH)4-1 = 13PrDiol(2)_B(OH)4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.025	Unknown	lgK:-0.70(2SF)	4	MGL	2118

2	13.05	0.025	Unknown	lgK:-0.92(2SF)	4	MGL	2118
3	25	0.025	Unknown	lgK:-0.96(2SF)	4	MGL	2118
4	34.98	0.025	Unknown	lgK:-1.25(3SF)	4	MGL	2118
5	0:34.98	0.025	Unknown	dH:-6.18(3SF)	4	MTD	2118

Reaction No. 59173

2<DGlucose> + B(OH)4-1 = B(OH)4-1_DGlucose(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.025	Unknown	lgK:2.95(3SF)	4	MGL	2118
2	13.05	0.025	Unknown	lgK:2.94(3SF)	4	MGL	2118
3	25	0.025	Unknown	lgK:2.94(3SF)	4	MGL	2118
4	34.98	0.025	Unknown	lgK:2.95(3SF)	4	MGL	2118
5	0:34.98	0.025	Unknown	dH:-0.149(3SF)	4	MTD	2118

Reaction No. 59174

Glycerol + B(OH)4-1 = B(OH)4-1_Glycerol

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.025	Unknown	lgK:1.36(3SF)	4	MGL	2118
2	13.05	0.025	Unknown	lgK:1.24(3SF)	4	MGL	2118
3	25	0.025	Unknown	lgK:1.15(3SF)	4	MGL	2118
4	34.98	0.025	Unknown	lgK:1.10(3SF)	4	MGL	2118
5	0:34.98	0.025	Unknown	dH:-2.85(3SF)	4	MTD	2118

Reaction No. 59175

2<Glycerol> + B(OH)4-1 = B(OH)4-1_Glycerol(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.025	Unknown	lgK:1.99(3SF)	4	MGL	2118
2	13.05	0.025	Unknown	lgK:1.84(3SF)	4	MGL	2118
3	25	0.025	Unknown	lgK:1.76(3SF)	4	MGL	2118
4	34.98	0.025	Unknown	lgK:1.62(3SF)	4	MGL	2118
5	0:34.98	0.025	Unknown	dH:-3.88(3SF)	4	MTD	2118

Reaction No. 59176

DMannitol + B(OH)4-1 = B(OH)4-1_DMannitol

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.025	Unknown	lgK:3.62(3SF)	4	MGL	2118
2	13.05	0.025	Unknown	lgK:3.36(3SF)	4	MGL	2118

3	25	0.025	Unknown	lgK:3.04(3SF)	4	MGL	2118
4	34.98	0.025	Unknown	lgK:2.90(3SF)	4	MGL	2118
5	0:34.98	0.025	Unknown	dH:-8.05(3SF)	4	MTD	2118

Reaction No. 59177

2<DMannitol> + B(OH)4-1 = B(OH)4-1_DMannitol(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.025	Unknown	lgK:5.42(3SF)	4	MGL	2118
2	13.05	0.025	Unknown	lgK:5.31(3SF)	4	MGL	2118
3	25	0.025	Unknown	lgK:5.14(3SF)	4	MGL	2118
4	34.98	0.025	Unknown	lgK:5.05(3SF)	4	MGL	2118
5	0:34.98	0.025	Unknown	dH:-4.45(3SF)	4	MTD	2118

Reaction No. 59178

Cat-2 + B(OH)4-1 = B(OH)4-1_Cat-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.025	Unknown	lgK:3.86(3SF)	4	MGL	2118
2	0	0.025	Unknown	lgK:4.09(3SF)	4	MGL	2118
3	13.05	0.025	Unknown	lgK:3.76(3SF)	4	MGL	2118
4	13.05	0.025	Unknown	lgK:4.01(3SF)	4	MGL	2118
5	25	0.025	Unknown	lgK:3.70(3SF)	4	MGL	2118
6	25	0.025	Unknown	lgK:3.90(3SF)	4	MGL	2118
7	34.98	0.025	Unknown	lgK:3.62(3SF)	4	MGL	2118
8	34.98	0.025	Unknown	lgK:3.81(3SF)	4	MGL	2118
9	0:34.98	0.025	Unknown	dH:-2.670(4SF)	4	MTD	2118
10	0:34.98	0.025	Unknown	dH:-3.190(4SF)	4	MTD	2118

Reaction No. 59179

2<Cat-2> + B(OH)4-1 = B(OH)4-1_Cat-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.025	Unknown	lgK:4.06(3SF)	4	MGL	2118
2	13.05	0.025	Unknown	lgK:4.09(3SF)	4	MGL	2118
3	25	0.025	Unknown	lgK:4.24(3SF)	4	MGL	2118
4	34.98	0.025	Unknown	lgK:4.34(3SF)	4	MGL	2118
5	0:34.98	0.025	Unknown	dH:3.77(3SF)	4	MTD	2118

Reaction No. 60964

B(OH)3 + H+1_Salicylic-2 = B+3_OH-1(2)_Salicylic-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5			lgK:1.37(3SF)	6	MSP	5987
2	15			lgK:1.19(3SF)	6	MSP	5987
3	20	0.1	KNO3	lgK:1.23(3SF)	6	MGL	5987
4	25	0.1	NaCl	lgK:1.03(3SF)	7	MSP	5987
5	25	0.1	NaClO4	lgK:1.04(3SF)	7	MSP	5987
6	25	0.1	Unknown	lgK:1.29(3SF)	4	EOD	8792
7	25			lgK:1.04(3SF)	6	MSP	5987

Reaction No. 60965							
B(OH)3 + H+1_2OH5ClBenz-2 = B+3_OH-1(2)_2OH5ClBenz-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:0.85(2SF)	6	MSP	5987

Reaction No. 60996							
B(OH)3 + H+1_2OH5BrBenz-2 = B+3_OH-1(2)_2OH5BrBenz-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:0.87(2SF)	6	MSP	5987

Reaction No. 61036							
B(OH)3 + H+1_5MeSal-2 = B+3_OH-1(2)_5MeSal-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaCl	lgK:1.05(3SF)	6	MSP	5987

Reaction No. 61045							
B(OH)3 + H+1_3NitrSal-2 = B+3_OH-1(2)_3NitrSal-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:0.38(2SF)	6	MSP	5987

Reaction No. 61046							
B(OH)3 + H+1_4NitrSal-2 = B+3_OH-1(2)_4NitrSal-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:0.76(2SF)	6	MSP	5987

Reaction No. 61052
B(OH)3 + H+1_5NitrSal-2 = B+3_OH-1(2)_5NitrSal-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:0.69(2SF)	7	MGL	5987
2	25			lgK:0.48(2SF)	6	MGL	5987

Reaction No. 61061							
B(OH)3 + H+1_6NitrSal-2 = B+3_OH-1(2)_6NitrSal-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:0.34(2SF)	6	MSP	5987

Reaction No. 61235							
B(OH)3 + H+1_5AmSal-2 = B+3_OH-1(2)_5AmSal-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:1.31(3SF)	6	MGL	5987

Reaction No. 61254							
B(OH)3 + H+1_1OH2Naphthoic-2 = B+3_OH-1(2)_1OH2Naphthoic-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:1.31(3SF)	6	MGL	5987

Reaction No. 61263							
B(OH)3 + H+1_2OH1Naphthoic-2 = B+3_OH-1(2)_2OH1Naphthoic-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:1.83(3SF)	6	MGL	5987

Reaction No. 61264							
B(OH)3 + H+1_2OH3Naphthoic-2 = B+3_OH-1(2)_2OH3Naphthoic-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:1.29(3SF)	6	MGL	5987

Reaction No. 65669							
Li+1 + B(OH)3 = Li+1_B(OH)3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.34(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:0.34(2SF)	4	CRV	552
3	25	0.7	Unknown	lgK:-0.05(1SF)	6	CRV	552

Reaction No. 65670
B(OH)3 + Na+1 = Na+1_B(OH)3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.2(0.05SD)	0	CRV	552
2	25	0.7	Unknown	lgK:-0.3(0.06SD)	0	CRV	552

Reaction No. 65671							
B(OH)3 + Mg+2 = Mg+2_B(OH)3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.5(0.09SD)	0	CRV	552
2	25	0.7	Unknown	lgK:0.9(0.2SD)	0	CRV	552
3	10:50	0	Inf. Dilution	dH:0.0(1SF)	0	CRV	552

Reaction No. 65672							
B(OH)3 + Ca+2 = Ca+2_B(OH)3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.8(0.04SD)	0	CRV	552
2	25	0.7	Unknown	lgK:1.1(0.03SD)	0	CRV	552
3	25	3	NaClO4	lgK:0.99(2SF)	0	CRV	552
4	10:50	0	Inf. Dilution	dH:1.(1SF)	0	CRV	552

Reaction No. 65673							
B(OH)3 + Sr+2 = Sr+2_B(OH)3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.55(3SF)	0	CRV	552
2	25	0.7	Unknown	lgK:0.54(2SF)	0	CRV	552
3	10:50	0	Inf. Dilution	dH:1.(1SF)	0	CRV	552

Reaction No. 65674							
Ba+2 + B(OH)3 = Ba+2_B(OH)3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.49(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:1.49(3SF)	4	CRV	552
3	10:50	0	Inf. Dilution	dH:1.(1SF)	4	CRV	552

Reaction No. 65675							
Cu+2 + B(OH)3 = Cu+2_B(OH)3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0	Unknown	lgK:7.13(3SF)	0	MSL	22560

2	25	0	Inf. Dilution	lgK:3.48(4SF)	1	EOR	
3	25	0.7	KNO3	lgK:3.48(3SF)	4	MSC	429
Note (3): Given in IUPAC database [05PeP_22560]							
4	25	0.7	Unknown	lgK:3.48(3SF)	6	CRV	552

Reaction No. 65676

Cu+2 + 2<B(OH)3> = Cu+2_B(OH)3(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0	Unknown	lgK:12.45(4SF)	0	MSL	22560
2	25	0	Inf. Dilution	lgK:6.13(4SF)	1	EOR	
3	25	0.7	KNO3	lgK:6.13(3SF)	4	MSC	429
Note (3): Given in IUPAC database [05PeP_22560]							
4	25	0.7	Unknown	lgK:6.13(3SF)	6	CRV	552

Reaction No. 65677

B(OH)3 + Zn+2 = Zn+2_B(OH)3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.7	Unknown	lgK:0.9(1SF)	0	CRV	552

Reaction No. 65678

Zn+2 + 2<B(OH)3> = Zn+2_B(OH)3(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.32(4SF)	1	EOR	
2	25	0.7	Unknown	lgK:3.32(3SF)	6	CRV	552

Reaction No. 65679

Cd+2 + B(OH)3 = Cd+2_B(OH)3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.42(4SF)	1	EOR	
2	25	0.7	Unknown	lgK:1.42(3SF)	6	CRV	552

Reaction No. 65680

Cd+2 + 2<B(OH)3> = Cd+2_B(OH)3(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.71(4SF)	1	EOR	
2	25	0.7	Unknown	lgK:2.71(3SF)	6	CRV	552

Reaction No. 65681							
Pb+2 + B(OH)3 = Pb+2_B(OH)3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.2(4SF)	1	EOR	
2	25	0.7	Unknown	lgK:2.2(2SF)	6	CRV	552

Reaction No. 65682							
Pb+2 + 2<B(OH)3> = Pb+2_B(OH)3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.41(4SF)	1	EOR	
2	25	0.7	Unknown	lgK:4.41(3SF)	6	CRV	552

Reaction No. 65683							
Fe+3 + B(OH)3 = Fe+3_B(OH)3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:8.5(2SF)	0	MSC	22560
2	23	0	Unknown	lgK:8.58(3SF)	0	MSC	22560
3	25	0	Inf. Dilution	lgK:1.0(4SF)	1	EOR	
4	25	0.68	Unknown	lgK:1.0(2SF)	2	MSP	413
Note (4): Given in IUPAC database [05PeP_22560]							
5	25	0.7	Unknown	lgK:6.85(3SF)	0	CRV	552

Reaction No. 65684							
Fe+3 + 2<B(OH)3> = Fe+3_B(OH)3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:15.6(3SF)	0	MSC	22560
2	23	0	Unknown	lgK:15.54(4SF)	0	MSC	22560
3	25	0	Inf. Dilution	lgK:2.0(4SF)	1	EOR	
4	25	0.68	Unknown	lgK:2.0(2SF)	2	MSP	413
Note (4): Given in IUPAC database [05PeP_22560]							
5	25	0.7	Unknown	lgK:13.0(3SF)	0	CRV	552

Reaction No. 65685							
B(OH)3 + Ag+1 = Ag+1_B(OH)3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.20(3SF)	4	CRV	552
2	25	3	NaClO4	lgK:0.45(2SF)	6	CRV	552

Reaction No. 66031							
H+1 + 3<B(OH)4-1> = H+1_B(OH)4-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.4(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:10.4(3SF)	4	ENS	6405

Reaction No. 66032							
4<H+1> + 5<B(OH)4-1> = H+1(4)_B(OH)4-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.8(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:38.8(3SF)	4	ENS	6405

Reaction No. 66034							
Ag+1_H+1_B(OH)4-1(2)_(s) = Ag+1 + H+1 + 2<B(OH)4-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-22.9(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:-22.9(3SF)	4	ENS	6405

Reaction No. 66290							
3<V(s)> + 4<B(s)> = V3B4(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:84.13(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:83.61(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:62.43(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:49.71(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:41.24(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-114.8(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-116.3(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-116.3(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-116.3(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-116.4(4SF)	4	MTD	6422

Reaction No. 66291							
3<V(s)> + 2<B(s)> = V3B2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:52.62(4SF)	4	ENS	6422

2	27	0	Inf. Dilution	lgK:52.29(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:39.07(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:31.13(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:25.84(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-71.78(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-72.6(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-72.6(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-72.6(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-72.7(3SF)	4	MTD	6422

Reaction No. 66292

V(s) + 2<B(s)> = VB2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:35.15(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:34.93(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:26.06(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:20.74(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:17.19(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-47.96(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-48.7(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-48.7(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-48.7(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-48.7(3SF)	4	MTD	6422

Reaction No. 66293

V(s) + B(s) = VB(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.98(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:23.83(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:17.80(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:14.18(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:11.77(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-32.71(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-33.1(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-33.1(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-33.1(3SF)	4	MTD	6422

10	327	0	Inf. Dilution	dH:-33.1(3SF)	4	MTD	6422

Reaction No. 66333							
U(s) + 12<B(s)> = UB12(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:70.57(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:70.14(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:52.89(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:42.58(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:35.72(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-96.27(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-94.9(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-94.5(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-94.2(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-94.1(3SF)	4	MTD	6422

Reaction No. 66334							
U(s) + 4<B(s)> = UB4(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.90(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:42.63(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:31.96(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:25.57(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:21.33(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-58.52(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-58.7(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-58.5(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-58.4(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-58.3(3SF)	4	MTD	6422

Reaction No. 66335							
U(s) + 2<B(s)> = UB2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.96(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:27.78(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:20.77(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:16.59(4SF)	4	ENS	6422

5	327	0	Inf. Dilution	lgK:13.81(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-38.14(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-38.6(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-38.4(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-38.2(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-38.0(3SF)	4	MTD	6422

Reaction No. 66365							
Ti(s) + 2<B(s)> = TiB2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:56.00(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:55.65(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:41.55(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:33.08(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:27.43(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-76.40(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-77.4(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-77.5(3SF)	3	MTD	6422
9	227	0	Inf. Dilution	dH:-77.5(3SF)	2	MTD	6422
10	327	0	Inf. Dilution	dH:-77.6(3SF)	0	MTD	6422

Reaction No. 66366							
Ti(s) + B(s) = TiB(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.98(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:27.80(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:20.82(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:16.62(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:13.82(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-38.17(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-38.3(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-38.4(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-38.4(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-38.4(3SF)	4	MTD	6422

Reaction No. 66503
B(s) + 1.5<Br2(l)> = B+3_Br-1(3)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:41.51(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:41.25(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:29.75(4SF)	2	ENS	6422
4	25	0	Inf. Dilution	dG:-56.62(4SF)	4	EFE	6422
5	25	0	Inf. Dilution	dH:-57.0(3SF)	2	MTD	6422
6	127	0	Inf. Dilution	dH:-66.6(3SF)	0	MTD	6422

Reaction No. 66504

B(s) + 0.5<N2(g)> = BN(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:40.03(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:39.76(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:28.67(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:22.01(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:17.57(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-228.4(4SF)kJ	3	CRV	6330
7	25	0	Inf. Dilution	dG:-54.61(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dG:-228.4(4SF)kJ	4	ENS	7381
9	25	0	Inf. Dilution	dH:-254.4(4SF)kJ	3	CRV	6330
10	25	0	Inf. Dilution	dH:-60.8(3SF)	4	MTD	6422
11	25	0	Inf. Dilution	dH:-254.4(4SF)kJ	4	ENS	7381
12	127	0	Inf. Dilution	dH:-60.9(3SF)	4	MTD	6422
13	227	0	Inf. Dilution	dH:-61.0(3SF)	4	MTD	6422
14	327	0	Inf. Dilution	dH:-61.0(3SF)	0	MTD	6422

Reaction No. 66505

2<B(s)> + 3<S(s)> = B+3(2)_S-2(3)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:43.38(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:43.10(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:32.14(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:25.41(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:20.88(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-59.18(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-60.3(3SF)	4	MTD	6422

8	127	0	Inf. Dilution	dH:-61.3(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-64.3(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-62.3(3SF)	4	MTD	6422

Reaction No. 66528

3<Be(s)> + 2<B(s)> + 3<O2(g)> = Be3B2O6(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:514.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:511.5(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:376.4(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:295.2(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:241.2(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-702.4(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-742.0(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-742.4(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-742.4(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-742.0(4SF)	4	MTD	6422

Reaction No. 66562

Ca(s) + 2<B(s)> + 2<O2(g)> = CaB2O4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:337.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:334.9(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:246.5(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:193.5(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:158.1(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-459.9(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-485.4(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-485.3(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-485.1(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-484.9(4SF)	4	MTD	6422

Reaction No. 66563

Ca(s) + 4<B(s)> + 3.5<O2(g)> = CaB4O7(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:554.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:551.2(4SF)	4	ENS	6422

3	127	0	Inf. Dilution	lgK:404.9(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:317.2(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:258.7(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-756.9(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-803.1(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-803.2(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-802.9(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-802.5(4SF)	4	MTD	6422

Reaction No. 66564

2<Ca(s)> + 2<B(s)> + 2.5<O2(g)> = Ca2B2O5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:454.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:452.0(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:332.9(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:261.6(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:214.0(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-620.6(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-653.5(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-653.3(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-652.9(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-652.4(4SF)	4	MTD	6422

Reaction No. 66565

3<Ca(s)> + 2<B(s)> + 3<O2(g)> = Ca3B2O6(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:571.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:567.4(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:418.2(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:328.7(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:269.0(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-779.12(5SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-819.6(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-819.3(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-818.8(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-818.2(4SF)	4	MTD	6422

Reaction No. 66618							
Ce(s) + 6<B(s)> = CeB6(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:57.59(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:57.42(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:42.69(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:33.86(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:27.96(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-78.84(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-81.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-80.8(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-80.8(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-81.0(3SF)	4	MTD	6422

Reaction No. 66635							
Co(s) + B(s) = CoB(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.21(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:16.11(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:12.01(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:9.552(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:7.910(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-22.12(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-22.5(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-22.5(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-22.5(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-22.6(3SF)	4	MTD	6422

Reaction No. 66636							
2<Co(s)> + B(s) = Co2B(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.(2SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:21.54(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:16.07(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:12.79(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:10.60(4SF)	4	ENS	6422

6	25	0	Inf. Dilution	dG:-29.57(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-30.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-30.0(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-30.0(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-30.1(3SF)	4	MTD	6422

Reaction No. 66653

Cr(s) + B(s) = CrB(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.49(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:13.41(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:10.13(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:8.173(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:6.868(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-18.40(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-18.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-18.0(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-17.9(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-17.9(3SF)	4	MTD	6422

Reaction No. 66654

Cr(s) + 2<B(s)> = CrB2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.68(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:16.58(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:12.49(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:10.05(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:8.414(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-22.76(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-22.5(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-22.4(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-22.4(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-22.4(3SF)	4	MTD	6422

Reaction No. 66709

Fe(s) + B(s) = FeB(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:12.18(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:12.10(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:9.030(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:7.208(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:5.999(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-16.61(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-17.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-16.7(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-16.6(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-16.6(3SF)	4	MTD	6422

Reaction No. 66710

2<Fe(s)> + B(s) = Fe2B(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.27(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:12.19(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:9.116(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:7.290(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:6.073(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-16.73(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-17.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-16.8(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-16.7(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-16.8(3SF)	4	MTD	6422

Reaction No. 66767

B(s) + 0.5<H2(g)> + O2(g) = HBO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:128.8(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:127.9(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:92.96(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:71.99(4SF)	3	ENS	6422
5	236	0	Inf. Dilution	lgK:70.51(4SF)	3	ENS	6422
6	25	0	Inf. Dilution	dG:-723.4(4SF)kJ	3	CRV	6330
7	25	0	Inf. Dilution	dG:-175.7(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-794.3(4SF)kJ	3	CRV	6330

9	25	0	Inf. Dilution	dH:-191.9(4SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-191.9(4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-191.8(4SF)	3	MTD	6422
12	236	0	Inf. Dilution	dH:-191.8(4SF)	3	MTD	6422

Reaction No. 66768

B(s) + 1.5<H2(g)> + 1.5<O2(g)> = H3BO3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:169.7(4SF)	2	ENS	6422
2	27	0	Inf. Dilution	lgK:168.5(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:120.9(4SF)	3	ENS	6422
4	171.1	0	Inf. Dilution	lgK:106.7(4SF)	3	ENS	6422
5	25	0	Inf. Dilution	dG:-968.9(4SF)kJ	2	CRV	6330
6	25	0	Inf. Dilution	dG:-231.5(4SF)	2	EFE	6422
7	25	0	Inf. Dilution	dG:-969.7(4SF)kJ	6	CRV	8746
8	25	0	Inf. Dilution	dG:-969.67(0.820SD)kJ	6	CRV	14923
9	25	0	Inf. Dilution	dG:-969.67(0.82SD)kJ	5	EOD	23679
10	25	0	Inf. Dilution	dG:-969.67(5SF)kJ	5	EOD	29520
11	25	0	Inf. Dilution	dH:-1094.3(5SF)kJ	5	CRV	6330
12	25	0	Inf. Dilution	dH:-261.5(4SF)	4	MTD	6422
13	25	0	Inf. Dilution	dH:-1094.8(0.8MD)kJ	6	CRV	8746
14	25	0	Inf. Dilution	dH:-1094.8(0.800SD)kJ	6	CRV	14923
15	25	0	Inf. Dilution	dH:-1094.8(0.8SD)kJ	5	EOD	23679
16	25	0	Inf. Dilution	dH:-1094.8(5SF)kJ	5	EOD	29520
17	127	0	Inf. Dilution	dH:-261.7(4SF)	3	MTD	6422
18	171.1	0	Inf. Dilution	dH:-261.8(4SF)	3	MTD	6422
19	25	0	Inf. Dilution	Cp:86.060(0.2SD)J/K	5	CRV	27292

Reaction No. 66934

B(s) + Li(s) + O2(g) = LiBO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:168.7(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:167.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:123.3(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:96.59(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:78.78(4SF)	0	ENS	6422

6	25	0	Inf. Dilution	dG:-976.1(4SF)kJ	3	CRV	6330
7	25	0	Inf. Dilution	dG:-230.2(4SF)	3	EFE	6422
8	25	0	Inf. Dilution	dH:-1032.2(5SF)kJ	3	CRV	6330
9	25	0	Inf. Dilution	dH:-243.6(4SF)	3	MTD	6422
10	127	0	Inf. Dilution	dH:-243.7(4SF)	3	MTD	6422
11	227	0	Inf. Dilution	dH:-244.4(4SF)	3	MTD	6422
12	327	0	Inf. Dilution	dH:-244.4(4SF)	0	MTD	6422

Reaction No. 66951

2<Li(s)> + 4<B(s)> + 3.5<O2(g)> = Li2B4O7(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:555.4(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:551.8(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:405.4(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:317.4(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:258.7(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-757.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-803.6(4SF)	3	MTD	6422
8	127	0	Inf. Dilution	dH:-804.1(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-806.1(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-806.6(4SF)	0	MTD	6422

Reaction No. 66952

2<Li(s)> + 6<B(s)> + 5<O2(g)> = Li2B6O10(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:767.8(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:762.8(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:560.0(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:438.3(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:357.1(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-1047.(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-1114.(4SF)	3	MTD	6422
8	127	0	Inf. Dilution	dH:-1113.(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-1114.(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-1114.(4SF)	0	MTD	6422

Reaction No. 66965

2<B(s)> + Mg(s) = MgB2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.69(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:15.59(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:11.58(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:9.169(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:7.549(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-21.40(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-22.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-22.0(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-22.2(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-22.3(3SF)	4	MTD	6422

Reaction No. 66966

4<B(s)> + Mg(s) = MgB4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.19(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:18.07(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:13.50(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:10.72(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:8.853(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-24.81(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-25.1(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-25.2(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-25.5(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-25.9(3SF)	4	MTD	6422

Reaction No. 66981

B(s) + Mn(s) = MnB(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.91(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:12.83(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:9.549(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:7.577(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:6.257(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-17.61(4SF)	4	EFE	6422

7	25	0	Inf. Dilution	dH:-18.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-18.0(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-18.1(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-18.2(3SF)	4	MTD	6422

Reaction No. 66982

2<B(s)> + Mn(s) = MnB2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.02(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:15.91(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:11.82(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:9.362(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:7.716(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-21.85(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-22.5(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-22.5(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-22.5(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-22.7(3SF)	4	MTD	6422

Reaction No. 67028

B(s) + Na(s) + O2(g) = NaBO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:161.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:160.0(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:117.5(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:91.96(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:74.92(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-920.7(4SF)kJ	3	CRV	6330
7	25	0	Inf. Dilution	dG:-219.7(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-977.0(4SF)kJ	3	CRV	6330
9	25	0	Inf. Dilution	dH:-233.2(4SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-233.9(4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-233.9(4SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-233.8(4SF)	0	MTD	6422

Reaction No. 67029

3<B(s)> + Na(s) + 2.5<O2(g)> = NaB3O5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:376.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:374.4(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:274.7(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:214.8(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:174.8(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-514.2(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-547.4(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-548.3(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-548.5(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-548.5(4SF)	4	MTD	6422

Reaction No. 67030

4<B(s)> + 2<Na(s)> + 3.5<O2(g)> = Na2B4O7(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:539.9(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:536.3(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:393.6(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:307.8(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:250.6(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-3096.0(5SF)kJ	3	CRV	6330
7	25	0	Inf. Dilution	dG:-736.5(4SF)	3	EFE	6422
8	25	0	Inf. Dilution	dH:-3291.1(5SF)kJ	3	CRV	6330
9	25	0	Inf. Dilution	dH:-783.2(4SF)	3	MTD	6422
10	127	0	Inf. Dilution	dH:-784.8(4SF)	3	MTD	6422
11	227	0	Inf. Dilution	dH:-785.2(4SF)	3	MTD	6422
12	327	0	Inf. Dilution	dH:-785.6(4SF)	0	MTD	6422

Reaction No. 67063

2<B(s)> + Nb(s) = NbB2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:43.44(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:43.17(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:32.24(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:25.68(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:21.31(4SF)	0	ENS	6422

6	25	0	Inf. Dilution	dG:-59.26(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-60.(2SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-60.0(3SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-61.(2SF)	4	CRV	22971
10	127	0	Inf. Dilution	dH:-60.0(3SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-60.0(3SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-60.0(3SF)	0	MTD	6422

Reaction No. 67099

B(s) + Ni(s) = NiB(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.30(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:17.19(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:12.82(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:10.18(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:8.423(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-23.60(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-24.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-24.1(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-24.1(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-24.3(3SF)	4	MTD	6422

Reaction No. 67100

3<B(s)> + 4<Ni(s)> = Ni4B3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:53.44(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:53.11(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:39.52(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:31.34(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:25.87(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-72.91(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-74.5(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-74.7(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-74.9(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-75.4(3SF)	4	MTD	6422

Reaction No. 67233

2<B(s)> + Ta(s) = TaB2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:36.19(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:35.97(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:26.86(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:21.40(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:17.75(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-49.37(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-50.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-50.0(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-50.0(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-50.0(3SF)	0	MTD	6422

Reaction No. 67246							
Pb(s) + 2<B(s)> + 2<O2(g)> = PbB2O4(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:254.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:252.4(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:184.6(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:144.0(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:117.0(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-346.6(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-372.0(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-371.9(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-371.6(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-371.0(4SF)	4	MTD	6422

Reaction No. 67247							
Pb(s) + 4<B(s)> + 3.5<O2(g)> = PbB4O7(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:467.3(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:464.2(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:339.8(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:265.2(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:215.5(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-637.4(4SF)	4	EFE	6422

7	25	0	Inf. Dilution	dH:-683.0(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-682.9(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-682.5(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-681.7(4SF)	4	MTD	6422

Reaction No. 67312

2<B(s)> + Hf(s) = HfB2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:58.21(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:57.84(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:43.22(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:34.46(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:28.62(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-79.41(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-80.30(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-80.24(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-80.18(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-80.14(4SF)	4	MTD	6422

Reaction No. 67372

B(s) + K(s) + O2(g) = KBO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:164.1(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:163.0(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:119.6(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:93.59(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:76.23(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-923.4(4SF)kJ	3	CRV	6330
7	25	0	Inf. Dilution	dG:-223.9(4SF)	3	EFE	6422
8	25	0	Inf. Dilution	dH:-981.6(4SF)kJ	3	CRV	6330
9	25	0	Inf. Dilution	dH:-237.8(4SF)	3	MTD	6422
10	127	0	Inf. Dilution	dH:-238.4(4SF)	3	MTD	6422
11	227	0	Inf. Dilution	dH:-238.4(4SF)	3	MTD	6422
12	327	0	Inf. Dilution	dH:-238.3(4SF)	0	MTD	6422

Reaction No. 67385

4<B(s)> + 2<K(s)> + 3.5<O2(g)> = K+1(2)_B4O7-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:549.5(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:545.9(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:400.6(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:313.3(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:255.1(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-749.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-796.9(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-798.8(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-799.2(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-799.5(4SF)	0	MTD	6422

Reaction No. 67386

6<B(s)> + 2<K(s)> + 5<O2(g)> = K2B6O10(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:762.7(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:757.7(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:555.9(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:434.7(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:353.8(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-1040.(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-1107.(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-1109.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-1110.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-1110.(4SF)	4	MTD	6422

Reaction No. 67387

8<B(s)> + 2<K(s)> + 6.5<O2(g)> = K2B8O13(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:978.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:971.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:712.7(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:557.2(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:453.5(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-1334.(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-1421.(4SF)	4	MTD	6422

8	127	0	Inf. Dilution	dH:-1423.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-1423.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-1423.(4SF)	4	MTD	6422

Reaction No. 67421

B(s) + Ca(s) + 0.5<H2(g)> + 2.5<O2(g)> + Si(s) = CaBSiO4OH(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:406.1(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:403.4(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:295.6(4SF)	3	ENS	6494
4	227	0	Inf. Dilution	lgK:230.8(4SF)	2	ENS	6494
5	327	0	Inf. Dilution	lgK:187.7(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-554.0(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-592.2(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-592.4(4SF)	3	MTD	6494
9	227	0	Inf. Dilution	dH:-592.3(4SF)	2	MTD	6494
10	327	0	Inf. Dilution	dH:-592.0(4SF)	0	MTD	6494

Reaction No. 67425

6.75<Al(s)> + B(s) + 0.375<H2(g)> + 9<O2(g)> + 3<Si(s)> = Al6.75Si3BO17.25(OH).75(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1501.(4SF)	3	ENS	6494
2	27	0	Inf. Dilution	lgK:1491.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:1095.(4SF)	3	ENS	6494
4	227	0	Inf. Dilution	lgK:856.7(4SF)	3	ENS	6494
5	327	0	Inf. Dilution	lgK:698.1(4SF)	3	ENS	6494
6	25	0	Inf. Dilution	dG:-2048.(4SF)	3	EFE	6494
7	25	0	Inf. Dilution	dH:-2177.(4SF)	3	MTD	6494
8	127	0	Inf. Dilution	dH:-2178.(4SF)	3	MTD	6494
9	227	0	Inf. Dilution	dH:-2178.(4SF)	3	MTD	6494
10	327	0	Inf. Dilution	dH:-2177.(4SF)	3	MTD	6494

Reaction No. 68822

BF4-1 + Na+1 = Na+1_BF4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		MeCN	lgK:1.7(2SF)	4	MCN	6714

Reaction No. 68823							
BF4-1 + K+1 = K+1_BF4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		MeCN	lgK:1.7(2SF)	4	MCN	6714

Reaction No. 68824							
BF4-1 + Rb+1 = Rb+1_BF4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		MeCN	lgK:1.8(2SF)	4	MCN	6714

Reaction No. 69112							
B+3 + H+1_Salicylic-2 = B+3_H+1_Salicylic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:7.131(4SF)	0	MSP	5987

Reaction No. 70176							
B(s) + 1.5<Cl2(g)> = BCl3(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-408.(3SF)kJ	4	EFE	7276

Reaction No. 70177							
B(s) + 1.5<Br2(l)> = BBr3(g)]							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-208.(3SF)kJ	4	EFE	7276

Reaction No. 70496							
2<B(s)> = B2(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:774.0(4SF)kJ	4	ENS	7381
2	25	0	Inf. Dilution	dH:830.5(4SF)kJ	4	ENS	7381

Reaction No. 70497							
2<B(s)> + 3<H2(g)> = B2H6(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:86.7(3SF)kJ	3	CRV	6330
2	25	0	Inf. Dilution	dG:86.7(3SF)kJ	4	ENS	7381
3	25	0	Inf. Dilution	dH:35.6(3SF)kJ	3	CRV	6330

4	25	0	Inf. Dilution	dH:35.6(3SF)kJ	4	ENS	7381

Reaction No. 70498

B(s) + 1.5<H2(g)> + 1.5<O2(g)> = B(OH)3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-231.6(4SF)	4	CRV	406
2	25	0	Inf. Dilution	dG:-968.75(5SF)kJ	0	ENS	7381
3	25	0	Inf. Dilution	dG:-969.3(4SF)kJ	6	CRV	8746
4	25	0	Inf. Dilution	dG:-231.54(5SF)	0	CRV	10174
5	25	0	Inf. Dilution	dG:-969.27(0.820SD)kJ	6	CRV	14923
6	25	0	Inf. Dilution	dG:-969.28(0.82SD)kJ	5	EOD	23679
7	25	0	Inf. Dilution	dG:-231.56(5SF)	2	EOD	29489
8	25	0	Inf. Dilution	dG:-969.27(5SF)kJ	5	EOD	29520
9	50	0	Inf. Dilution	dG:-232.48(5SF)	0	CRV	10174
10	75	0	Inf. Dilution	dG:-233.44(5SF)	0	CRV	10174
11	100	0	Inf. Dilution	dG:-234.43(5SF)	0	CRV	10174
12	125	0	Inf. Dilution	dG:-235.43(5SF)	0	CRV	10174
13	150	0	Inf. Dilution	dG:-236.46(5SF)	0	CRV	10174
14	175	0	Inf. Dilution	dG:-237.51(5SF)	0	CRV	10174
15	200	0	Inf. Dilution	dG:-238.59(5SF)	0	CRV	10174
16	225	0	Inf. Dilution	dG:-239.68(5SF)	0	CRV	10174
17	250	0	Inf. Dilution	dG:-240.80(5SF)	0	CRV	10174
18	300	0	Inf. Dilution	dG:-243.12(5SF)	0	CRV	10174
19	25	0	Inf. Dilution	dH:-1072.30(6SF)kJ	4	ENS	7381
20	25	0	Inf. Dilution	dH:-1072.8(0.8MD)kJ	6	CRV	8746
21	25	0	Inf. Dilution	dH:-1072.8(0.800SD)kJ	6	CRV	14923
22	25	0	Inf. Dilution	dH:-1072.80(6SF)kJ	5	EOD	23679
23	25	0	Inf. Dilution	dH:-256.29(5SF)	2	EOD	29489
24	25	0	Inf. Dilution	dH:-1072.8(5SF)kJ	5	EOD	29520

Reaction No. 70499

B(s) + O2(g) + e-1 = BO2-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-162.24(5SF)	4	CRV	6488
2	25	0	Inf. Dilution	dG:-678.89(5SF)kJ	4	ENS	7381
3	25	0	Inf. Dilution	dH:-184.6(4SF)	4	CRV	6488

4	25	0	Inf. Dilution	dH:-772.37(5SF)kJ	4	ENS	7381

Reaction No. 71171							
H3BO3(s) + 3<H+1> + 3<e-1> = B(s) + 3<H2O>							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:-44.8(3SF)	1	EOR	
2	25	3	Inert	lgK:-45.9(3SF)	1	EOR	
3	25	0	Inf. Dilution	E:-0.890(3SF)	5	CRV	6330
4	25	0	Inf. Dilution	E:-0.889(3SF)	5	CRV	7761
5	25	0	Inf. Dilution	dH:237.0(4SF)kJ	5	CRV	7761

Reaction No. 71683							
B(OH)4-1 + 3<e-1> = B(s) + 4<OH-1>							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	E:-1.811(4SF)	3	CRV	7761
2	25	0	Inf. Dilution	E:-1.811(4SF)	4	CRV	22551
3	25	0	Inf. Dilution	dH:421.1(4SF)kJ	3	CRV	7761

Reaction No. 71707							
2<H3BO3(s)> + 12<H+1> + 12<e-1> = B2H6(g) + 6<H2O>							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:-105.6(4SF)	1	EOR	
2	25	3	Inert	lgK:-107.6(4SF)	1	EOR	
3	25	0	Inf. Dilution	E:-0.520(3SF)	0	CRV	7761
4	25	0	Inf. Dilution	dH:509.9(4SF)kJ	3	CRV	7761

Reaction No. 71708							
2<B(s)> + 6<e-1> + 6<H+1> = B2H6(g)							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	E:-0.150(3SF)	5	CRV	7761
2	25	0	Inf. Dilution	E:-0.15(2SF)	4	CRV	22551
3	25	0	Inf. Dilution	dH:35.7(3SF)kJ	5	CRV	7761

Reaction No. 71709							
B(s) + 4<H+1> + 5<e-1> = BH4-1							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:-19.9(3SF)	1	ELC	

2	25	0	Inf. Dilution	E:-0.237(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:48.2(3SF)kJ	2	CRV	7761

Reaction No. 71710

B2H6(g) + 2<H+1> + 4<e-1> = 2<BH4-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-24.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.368(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:60.9(3SF)kJ	2	CRV	7761

Reaction No. 71711

B(OH)3 + 3<e-1> + 3<H+1> = B(s) + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	Inert	lgK:-45.9(3SF)	1	EOR	
2	25	0	Inf. Dilution	E:-0.889(3SF)	5	CRV	7761
3	25	0	Inf. Dilution	E:-0.89(2SF)	4	CRV	22551
4	25	0	Inf. Dilution	dH:214.9(4SF)kJ	5	CRV	7761

Reaction No. 71712

B(OH)3 + 7<H+1> + 8<e-1> = BH4-1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	Inert	lgK:-66.2(3SF)	1	EOR	
2	25	0	Inf. Dilution	E:-0.481(3SF)	5	CRV	7761
3	25	0	Inf. Dilution	dH:262.7(4SF)kJ	5	CRV	7761

Reaction No. 71713

H3BO3(s) + 7<H+1> + 8<e-1> = BH4-1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.482(3SF)	5	CRV	7761
2	25	0	Inf. Dilution	dH:285.3(4SF)kJ	5	CRV	7761

Reaction No. 71714

2<B(OH)3> + 12<e-1> + 12<H+1> = B2H6(g) + 6<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-105.5(4SF)	1	EOR	
2	25	3	Inert	lgK:-107.6(4SF)	1	EOR	
3	25	0	Inf. Dilution	E:-0.519(3SF)	0	CRV	7761

4	25	0	Inf. Dilution	E:-0.519(3SF)	0	CRV	22551
5	25	0	Inf. Dilution	dH:464.9(4SF)kJ	2	CRV	7761

Reaction No. 72355

B(OH)3 + H+1_Cat-2 = B+3_OH-1(2)_Cat-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:4.2(2SF)	4	EOD	8792

Reaction No. 72356

B(OH)3 + H+1_Tartaric-2 = B+3_OH-1(2)_Tartaric-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:0.9(1SF)	4	EOD	8792

Reaction No. 72357

B(OH)3 + H+1_Oxalic-2 = B+3_OH-1(2)_Oxalic-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:0.35(2SF)	4	EOD	8792

Reaction No. 72358

B(OH)3 + H+1_CO3-2 = B+3_OH-1(2)_CO3-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.7	NaCl/m	lgK:0.415(0.23SD)	6	MSP	8792

Reaction No. 72413

Cu+2 + 3<B(OH)4-1> = Cu+2_B(OH)4-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.5(2SF)	1	EES	
2	25	0	Inf. Dilution	lgK:8.5(4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:15.17(4SF)	0	EOD	406

Note (3): Data from Shchigol, RJIC, 65, 10, 1142 (far too high - see B110 & B120)

Reaction No. 73063

B+3 + 3<e-1> = B(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.89(2SF)	0	CRV	22551

Note (1): Implied same as B(OH)3 + 3<e-1> + 3<H+1> = B(s) + 3<H2O>

Reaction No. 73758

B(OH)3 + OH-1 = B(OH)4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	Na+1 salt	lgK:5.22(0.01MD)	6	MHY	22982
2	49.59	0.129	KCl/m	lgK:4.224(4SF)	6	MHY	409
3	49.79	0.266	KCl/m	lgK:4.234(4SF)	6	MHY	409
4	49.94	0.51	KCl/m	lgK:4.270(4SF)	6	MHY	409
5	50.14	0.129	KCl/m	lgK:4.201(4SF)	6	MHY	409
6	50.19	0.266	KCl/m	lgK:4.226(4SF)	6	MHY	409
7	50.34	0.51	KCl/m	lgK:4.260(4SF)	6	MHY	409
8	50.54	1.02	KCl/m	lgK:4.298(4SF)	6	MHY	409
9	99.55	0.129	KCl/m	lgK:3.340(4SF)	6	MHY	409
10	99.6	0.266	KCl/m	lgK:3.359(4SF)	6	MHY	409
11	99.6	0.51	KCl/m	lgK:3.387(4SF)	6	MHY	409
12	99.6	0.51	KCl/m	lgK:3.389(4SF)	6	MHY	409
13	99.65	0.266	KCl/m	lgK:3.359(4SF)	6	MHY	409
14	99.67	0.129	KCl/m	lgK:3.341(4SF)	6	MHY	409
15	99.8	1.02	KCl/m	lgK:3.423(4SF)	6	MHY	409
16	100.45	1.02	KCl/m	lgK:3.422(4SF)	6	MHY	409
17	149.42	0.129	KCl/m	lgK:2.721(4SF)	6	MHY	409
18	149.99	0.266	KCl/m	lgK:2.730(4SF)	6	MHY	409
19	150.12	0.266	KCl/m	lgK:2.730(4SF)	6	MHY	409
20	150.12	0.51	KCl/m	lgK:2.759(4SF)	6	MHY	409
21	150.14	0.129	KCl/m	lgK:2.713(4SF)	6	MHY	409
22	150.62	1.02	KCl/m	lgK:2.782(4SF)	6	MHY	409
23	150.87	1.02	KCl/m	lgK:2.791(4SF)	6	MHY	409
24	198.23	0.129	KCl/m	lgK:2.294(4SF)	6	MHY	409
25	199.88	0.51	KCl/m	lgK:2.323(4SF)	6	MHY	409
26	200.03	1.02	KCl/m	lgK:2.344(4SF)	6	MHY	409
27	200.23	0.51	KCl/m	lgK:2.320(4SF)	6	MHY	409
28	200.33	0.266	KCl/m	lgK:2.291(4SF)	6	MHY	409
29	200.43	0.266	KCl/m	lgK:2.293(4SF)	6	MHY	409
30	200.53	1.02	KCl/m	lgK:2.344(4SF)	6	MHY	409
31	250.28	1.02	KCl/m	lgK:2.022(4SF)	6	MHY	409
32	250.43	0.129	KCl/m	lgK:1.989(4SF)	6	MHY	409
33	250.48	0.51	KCl/m	lgK:2.023(4SF)	6	MHY	409
34	250.48	1.02	KCl/m	lgK:2.034(4SF)	6	MHY	409

35	250.78	0.266	KCl/m	lgK:2.000(4SF)	6	MHY	409
36	251.08	0.266	KCl/m	lgK:1.991(4SF)	6	MHY	409
37	251.58	0.129	KCl/m	lgK:2.006(4SF)	6	MHY	409
38	290.03	1.02	KCl/m	lgK:1.867(4SF)	6	MHY	409
39	290.08	0.129	KCl/m	lgK:1.830(4SF)	6	MHY	409
40	290.28	0.51	KCl/m	lgK:1.865(4SF)	6	MHY	409
41	290.43	0.266	KCl/m	lgK:1.854(4SF)	6	MHY	409
42	0	0	Inf. Dilution/m	dH:-10.25(0.13MD)	5	MTD	409
43	25	0	Inf. Dilution/m	dH:-10.12(0.08MD)	5	MTD	409
44	50	0	Inf. Dilution/m	dH:-9.92(0.05MD)	5	MTD	409
45	75	0	Inf. Dilution/m	dH:-9.65(0.05MD)	5	MTD	409
46	100	0	Inf. Dilution/m	dH:-9.31(0.05MD)	5	MTD	409
47	125	0	Inf. Dilution/m	dH:-8.90(0.06MD)	5	MTD	409
48	150	0	Inf. Dilution/m	dH:-8.42(0.06MD)	5	MTD	409
49	175	0	Inf. Dilution/m	dH:-7.88(0.06MD)	5	MTD	409
50	200	0	Inf. Dilution/m	dH:-7.26(0.08MD)	5	MTD	409
51	225	0	Inf. Dilution/m	dH:-6.58(0.11MD)	5	MTD	409
52	250	0	Inf. Dilution/m	dH:-5.82(0.15MD)	5	MTD	409
53	275	0	Inf. Dilution/m	dH:-5.00(0.22MD)	5	MTD	409
54	300	0	Inf. Dilution/m	dH:-4.11(0.29MD)	0	MTD	409
55	0	0	Inf. Dilution/m	Cp:3.9(2.6MD)	5	MTD	409
56	25	0	Inf. Dilution/m	Cp:6.7(2.1MD)	5	MTD	409
57	50	0	Inf. Dilution/m	Cp:9.4(1.6MD)	5	MTD	409
58	75	0	Inf. Dilution/m	Cp:12.2(1.2MD)	5	MTD	409
59	100	0	Inf. Dilution/m	Cp:15.0(0.8MD)	5	MTD	409
60	125	0	Inf. Dilution/m	Cp:17.7(0.7MD)	5	MTD	409
61	150	0	Inf. Dilution/m	Cp:20.5(0.8MD)	5	MTD	409
62	175	0	Inf. Dilution/m	Cp:23.3(1.2MD)	5	MTD	409
63	200	0	Inf. Dilution/m	Cp:26.0(1.6MD)	5	MTD	409
64	225	0	Inf. Dilution/m	Cp:28.8(2.1MD)	5	MTD	409
65	250	0	Inf. Dilution/m	Cp:31.5(2.6MD)	5	MTD	409
66	275	0	Inf. Dilution/m	Cp:34.3(3.0MD)	5	MTD	409
67	300	0	Inf. Dilution/m	Cp:37.7(3.5MD)	4	MTD	409

Reaction No. 74315
B4O7-2 + 24<e-1> + 26<H+1> = 2<B2H6(g)> + 7<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.483(3SF)	4	CRV	22551

Reaction No. 74316

BO2-1 + 8<e-1> + 6<H2O> = BH4-1 + 8<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.241(4SF)	4	CRV	22551

Reaction No. 74317

3<B(OH)3> = B3O3(OH)4-1 + H+1 + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.85(3SF)	3	CRV	22551
2	25	0.1	NaClO4	lgK:-7.29(0.015MD)	5	MHY	427
3	25	3	NaClO4	lgK:-6.91(0.028MD)	5	MHY	427

Reaction No. 74659

5<B(OH)3> = B5O6(OH)4-1 + H+1 + 5<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:-6.77(0.102MD)	4	MHY	427
2	25	3	NaClO4	lgK:-6.62(0.156MD)	4	MHY	427

Reaction No. 74669

4<B(OH)4-1> + 2<H+1> = B4O7-2 + 9<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.9(3SF)	1	ELC	

Reaction No. 75142

4<B(OH)4-1> = B4O5(OH)4-2 + 2<OH-1> + 5<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.1(2SF)	1	ELC	
2	25	3	Na+1 salt	lgK:-7.170(0.039MD)	0	MHY	22982

Reaction No. 75143

3<B(OH)4-1> = B3O3(OH)5-2 + OH-1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.805(4SF)	2	EDH	
2	25	3	Na+1 salt	lgK:-4.64(0.19SD)	3	MHY	22982

Note (2): Species refuted by Bassett, Geochim. Cosmochim. Acta, 1980, 44, 1151

Reaction No. 75144

3<B(OH)3> + 2<OH-1> = B3O3(OH)5-2 + 3<H2O>

Note: Species refuted by Bassett, Geochim. Cosmochim. Acta, 1980, 44, 1151

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.5(2SF)	1	ELC	
2	25	3	Na+1 salt	lgK:11.29(0.19MD)	0	MHY	22982

Note (2): Dup. 3<B(OH)4-1> = B3O3(OH)5-2 + OH-1 + 3<H2O>

Reaction No. 75442

B(s) + 1.5<H2(g)> + 1.5<O2(g)> = H3BO3(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-931.88(5SF)kJ	5	EOD	23679
2	25	0	Inf. Dilution	dH:-1003.90(0.82SD)kJ	5	EOD	23679

Reaction No. 75443

H3BO3(s) = B(OH)3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:0.392(0.020SD)kJ	6	EOD	23679
2	25	0	Inf. Dilution	dH:22.00(0.05SD)kJ	6	EOD	23679

Reaction No. 75444

H3BO3(s) = H3BO3(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:37.79(4SF)kJ	6	EOD	23679
2	25	0	Inf. Dilution	dH:90.9(0.2SD)kJ	6	EOD	23679

Reaction No. 75445

H3BO3(g) = B(OH)3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-37.40(4SF)kJ	6	EOD	23679
2	25	0	Inf. Dilution	dH:-68.9(0.2SD)kJ	6	EOD	23679

Reaction No. 75564

B(s) + 2<F2(g)> + Na(s) = Na+1_BF4-1_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	dG:-1750.1(5SF)kJ	5	CRV	6330
2	25	0	Inf. Dilution	dH:-1844.7(5SF)kJ	5	CRV	6330

Reaction No. 75565

B(s) + 0.5<F2(g)> = BF(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-149.8(4SF)kJ	5	CRV	6330
2	25	0	Inf. Dilution	dH:-122.2(4SF)kJ	5	CRV	6330

Reaction No. 75569

B(s) + 2<H2(g)> + K(s) = K+1_BH4-1_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-160.3(4SF)kJ	5	CRV	6330
2	25	0	Inf. Dilution	dH:-227.4(4SF)kJ	5	CRV	6330

Reaction No. 75570

B(s) + 2<H2(g)> + Li(s) = Li+1_BH4-1_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-125.0(4SF)kJ	5	CRV	6330
2	25	0	Inf. Dilution	dH:-190.8(4SF)kJ	5	CRV	6330

Reaction No. 75571

B(s) + 2<H2(g)> + Na(s) = Na+1_BH4-1_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-123.9(4SF)kJ	5	CRV	6330
2	25	0	Inf. Dilution	dH:-188.6(4SF)kJ	5	CRV	6330

Reaction No. 75572

B(s) + 1.5<I2(s)> = BI3(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:20.7(3SF)kJ	5	CRV	6330
2	25	0	Inf. Dilution	dH:71.1(3SF)kJ	5	CRV	6330

Reaction No. 75573

B(s) + 0.5<N2(g)> = BN(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:614.5(4SF)kJ	3	CRV	6330

2	25	0	Inf. Dilution	dH:647.5(4SF)kJ	3	CRV	6330

Reaction No. 75574

B(s) + 0.5<O2(g)> = BO(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-4.0(2SF)kJ	3	CRV	6330
2	25	0	Inf. Dilution	dH:25.0(3SF)kJ	3	CRV	6330

Reaction No. 75575

B(s) + O2(g) = BO2(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-305.9(4SF)kJ	3	CRV	6330
2	25	0	Inf. Dilution	dH:-300.4(4SF)kJ	3	CRV	6330

Reaction No. 75576

B(s) + O2(g) + Rb(s) = RbBO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-913.0(4SF)kJ	3	CRV	6330
2	25	0	Inf. Dilution	dH:-971.0(4SF)kJ	3	CRV	6330

Reaction No. 75577

B(s) + S(s) = BS(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:288.8(4SF)kJ	3	CRV	6330
2	25	0	Inf. Dilution	dH:342.0(4SF)kJ	3	CRV	6330

Reaction No. 75578

2<B(s)> + 2<Cl2(g)> = B2Cl4(l)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-464.8(4SF)kJ	3	CRV	6330
2	25	0	Inf. Dilution	dH:-523.0(4SF)kJ	3	CRV	6330

Reaction No. 75579

2<B(s)> + 2<Cl2(g)> = B2Cl4(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-460.6(4SF)kJ	3	CRV	6330
2	25	0	Inf. Dilution	dH:-490.4(4SF)kJ	3	CRV	6330

Reaction No. 75580

2<B(s)> + 2<F2(g)> = B2F4(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1410.4(5SF)kJ	3	CRV	6330
2	25	0	Inf. Dilution	dH:-1440.1(5SF)kJ	3	CRV	6330

Reaction No. 75581

2<B(s)> + 1.5<O2(g)> = B2O3(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-832.0(4SF)kJ	3	CRV	6330
2	25	0	Inf. Dilution	dH:-843.8(4SF)kJ	3	CRV	6330

Reaction No. 75639

Cu+2_B(OH)3 + B(OH)3 = Cu+2_B(OH)3(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0	Unknown	lgK:5.32(3SF)	0	MSL	22560
2	25	0.7	KNO3	lgK:2.65(3SF)	4	MSC	429

Note (2): Given in IUPAC database [05PeP_22560]

Reaction No. 75640

Cu+2_B(OH)3(2) + B(OH)3 = Cu+2_B(OH)3(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0	Unknown	lgK:2.72(3SF)	0	MSL	22560

Reaction No. 75641

3<B(OH)3> + Cu+2 = Cu+2_B(OH)3(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0	Unknown	lgK:15.5(3SF)	0	MSL	22560

Note (1): Ranges from 15.2 - 15.7

Reaction No. 76497

Me4N+1_Ph4B-1_(s) = Me4N+1 + Ph4B-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:61.92(4SF)kJ	7	CRV	22677

Reaction No. 76502

2<B(s)> + Ca(s) + 4<O2(g)> + 2<Si(s)> = CaB2Si2O8(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:644.2(4SF)	3	CRV	6494
2	25	0	Inf. Dilution	dG:-3677.0(5SF)kJ	3	CRV	6494
3	25	0	Inf. Dilution	dH:-3902.8(5SF)kJ	3	CRV	6494

Reaction No. 77283

2<B(s)> + 4<H2(g)> + Ni(s) + 4<O2(g)> = Ni+2_B(OH)3(2)_OH-1(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2525.6(5SF)kJ	4	CRV	9015

Reaction No. 77513

BrO-1 + B(OH)3 = B(OH)3_BrO-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.14(4SF)	1	ELC	

Reaction No. 77675

ClO-1 + B(OH)3 = B(OH)3_ClO-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.409(4SF)	1	ELC	

Reaction No. 78203

2<OH-1> + Ni+2 + 2<B(OH)3> = Ni+2_B(OH)3(2)_OH-1(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:39.81(4SF)	1	ELC	

Reaction No. 78457

-1<H+1> + B(OH)3 = H+1(-1)_B(OH)3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.407(4SF)	0	ELC	

Reaction No. 79200

Pb+2 + 3<B(OH)4-1> = Pb+2_B(OH)4-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.85(4SF)	1	ELC	

Reaction No. 79201

Ni+2 + 3<B(OH)4-1> = Ni+2_B(OH)4-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:8.231(4SF)	1	ELC	

Reaction No. 79202

Co+2 + 4<B(OH)4-1> = Co+2_B(OH)4-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.135(4SF)	1	ELC	

Reaction No. 79203

Cd+2 + 4<B(OH)4-1> = Cd+2_B(OH)4-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.16(4SF)	1	ELC	

Reaction No. 79204

Zn+2 + 4<B(OH)4-1> = Zn+2_B(OH)4-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.32(4SF)	1	ELC	

Reaction No. 79426

Na+1 + BF4-1 = Na+1_BF4-1_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.1982(4SF)	1	ELC	

Reaction No. 79460

B(s) + 2<H2(g)> + e-1 = BH4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:114.35(5SF)kJ	2	CRV	7421

Reaction No. 80038

3<B(OH)3> + Ni+2 = Ni+2_B3O4(OH)3-2 + 2<H+1> + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-11.50(0.05SD)	5	MSL	29899
2	50	0	Inf. Dilution	lgK:-10.50(0.06SD)	5	MSL	29899
3	70	0	Inf. Dilution	lgK:-10.00(0.05SD)	5	MSL	29899
4	25	0	Inf. Dilution	dG:65.4(3SF)kJ	5	EFE	29899
5	25	0	Inf. Dilution	dH:65.6(3SF)kJ	5	EFE	29899

Reaction No. 80964

Ag+1_Ph4B-1_(s) = Ag+1 + Ph4B-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-17.2(3SF)	2	MIX	32205

Reaction No. 80968							
Ph4As+1_Ph4B-1_(s) = Ph4As+1 + Ph4B-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.30(3SF)	4	MSL	32204

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CMN	Several experimental values (crit.)
CRV	Critical review
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
ELC	Linear Combination of Reactions
ENS	Not specified
EOD	Calculated from data of others

EOR	Other Reaction Constants
ESC	Standard conditions anchor
MCL	Calorimetry
MCN	Conductivity
MCO	Colorimetry (often for measuring pH)
MDG	Combination of Thermodynamic Data
MDM	Emf by differential method (buffer capacity)
MEF	Electromotoric force, not specified
MGL	Glass electrode
MHY	Emf with hydrogen electrode
MIS	Ion selective electrode
MIX	Ion exchange
MMS	Magnetic susceptibility
MOR	Optical Rotary Dispersion
MPH	pH method, not specified
MPK	Protonation constant shift
MPL	Polarography
MPS	Potentiometry and Spectrophotometry
MPT	Potentiometry
MSC	Miscellaneous
MSL	Solubility
MSP	Spectroscopy
MSV	Anodic Stripping Voltammetry
MTD	Temperature difference

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